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Constraining sequence stratigraphy in north Australian basins: SHRIMP U–Pb zircon geochronology between Mt Isa and McArthur River*

R. W. PAGE, M. J. JACKSON AND A. A. KRASSAY

Australian Geological Survey Organisation, GPO Box 378, Canberra, ACT 2601, Australia.

Fifty-five new SHRIMP U–Pb zircon ages from samples of northern Australian 'basement' and its overlying Proterozoic successions are used to refine and, in places, significantly change previous lithostratigraphic correlations. In conjunction with sequence-stratigraphic studies, the 1800–1580 Ma rock record between Mt Isa and the Roper River is now classified into three superbasin phases—the Leichhardt, Calvert and Isa. These three major depositional episodes are separated by ~20 million years gaps. The Isa Superbasin can be further subdivided into seven supersequences each 10–15 million years in duration. Gaps in the geological record between these supersequences are variable; they approach several million years in basin-margin positions, but are much smaller in the depocentres. Arguments based on field setting, petrography, zircon morphology, and U–Pb systematics are used to interpret these U–Pb zircon ages and in most cases to demonstrate that the ages obtained are depositional. In some instances, zircon crystals are reworked and give maximum depositional ages. These give useful provenance information as they fingerprint the source(s) of basin fill. Six new 'Barramundi' basement ages (around 1850 Ma) were obtained from crystalline units in the Murphy Inlier (Nicholson Granite and Cliffdale Volcanics), the Urapunga Tectonic Ridge ('Mt Reid Volcanics' and 'Urapunga Granite'), and the central McArthur Basin (Scrutton Volcanics). New ages were also obtained from units assigned to the Calvert Superbasin (*ca* 1740–1690 Ma). SHRIMP results show that the Wollogorang Formation is not one continuous unit, but two different sequences, one deposited around 1730 Ma and a younger unit deposited around 1722 Ma. Further documentation is given of a regional 1725 Ma felsic event adjacent to the Murphy Inlier (Peters Creek Volcanics and Packsaddle Microgranite) and in the Carrara Range. A younger *ca* 1710 Ma felsic event is indicated in the southwestern McArthur Basin (Tanumbirini Rhyolite and overlying Nyanantu Formation). Four of the seven supersequences in the Isa Superbasin (*ca* 1670–1580 Ma) are reasonably well-constrained by the new SHRIMP results: the Gun Supersequence (*ca* 1670–1655 Ma) by Paradise Creek Formation, Moondarra Siltstone, Breakaway Shale and Urquhart Shale ages grouped between 1668 and 1652 Ma; the Loretta Supersequence (*ca* 1655–1645 Ma) by results from the Lady Loretta Formation, Walford Dolomite, the upper part of the Mallapunyah Formation and the Tatoola Sandstone between *ca* 1653 and 1647 Ma; the River Supersequence (*ca* 1645–1630 Ma) by ages from the Teena Dolomite, Mt Les and Riversleigh Siltstones, and Barney Creek, Lynott, St Vidgeon and Nagi Formations clustering around 1640 Ma; and the Term Supersequence (*ca* 1630–1615 Ma) by ages from the Stretton Sandstone, lower Doomadgee Formation and lower part of the Lawn Hill Formation, mostly around 1630–1620 Ma. The next two younger supersequences are less well-constrained geochronologically, but comprise the Lawn Supersequence (*ca* 1615–1600 Ma) with ages from the lower Balbirini Dolomite, and lower Doomadgee, Amos and middle Lawn Hill Formations, clustered around 1615–1610 Ma; and the Wide Supersequence (*ca* 1600–1585 Ma) with only two ages around 1590 Ma, one from the upper Balbirini Dolomite and the other from the upper Lawn Hill Formation. The Doom Supersequence (<1585 Ma) at the top of the Isa Superbasin is essentially unconstrained. The integration of high-precision SHRIMP dating from continuously analysed stratigraphic sections, within a sequence stratigraphic context, provides an enhanced chronostratigraphic framework leading to more reliable interpretations of basin architecture and evolution.

KEY WORDS: geochronology, Lawn Hill Platform, McArthur Basin, Mt Isa Inlier, Murphy Inlier, sequence stratigraphy, U–Pb zircon dating, Urapunga Inlier.

INTRODUCTION

There have been earlier endeavours using U–Pb zircon geochronology (Page 1981; Page & Sweet 1998) to date various stratigraphic components in the NABRE study area. Such studies established an age framework that broadly defined the chronology of various lithostratigraphic depositional packages across the Lawn Hill Platform and McArthur Basin.

The present geochronological study was integrated with the sequence-stratigraphic studies that included 46 km

*Tables 2–48 [indicated by an asterisk (*) in the text and listed at the end of the paper] are Supplementary Papers lodged with the National Library of Australia (Manuscript Section); copies may be obtained from the Business Manager, Geological Society of Australia.

of outcrop and 20 km of drillcore stratigraphic data (Southgate *et al.* 1999). The new ages reported here define and constrain particular sequence-stratigraphic links within and between the major basins. A particular benefit of this multidisciplinary approach was the ability to identify otherwise unrecognised tuffaceous layers signalled by peaks in gamma-ray logs, especially at some maximum flooding surfaces. There is a better chance of preservation of thin air-fall tuffs/reworked tuffs in these deeper water environments that are relatively starved of clastic input.

Benefits and pitfalls of using U–Pb ages to date zircon crystallisation (and hopefully depositional ages) in these commonly high-K tuffs have been previously outlined (Page 1981; Page & Sweet 1998). A stratigraphically useful tuff age comes from a zircon suite crystallised in explosive tuffaceous volcanism that is contemporary with deposition. Although tuff layers are commonly reworked, the time interval involved in reworking prior to burial and preservation is not generally significant and is probably within analytical uncertainty for Proterozoic age determinations. Rigorously, the U–Pb crystallisation age of zircon in a volcaniclastic rock is a maximum age for deposition. It can be used as a true depositional age only when there are sufficient coherent data and when there is a sequence of such ages that can be stratigraphically validated—that is, predictable age superposition. Tuffaceous rocks used in this study are generally of subgreenschist facies, and metamorphic zircon overprinting problems are not an issue.

Detailed gamma-ray, lithofacies, and sequence-stratigraphic studies in part of the NABRE study area (Figure 1) have refined the long-established lithostratigraphy. As individual U–Pb zircon tuff ages can seldom be reliably interpreted in isolation, this new and improved genetic packaging provides the basic controls that underpin the interpretation of the new depositional ages presented in

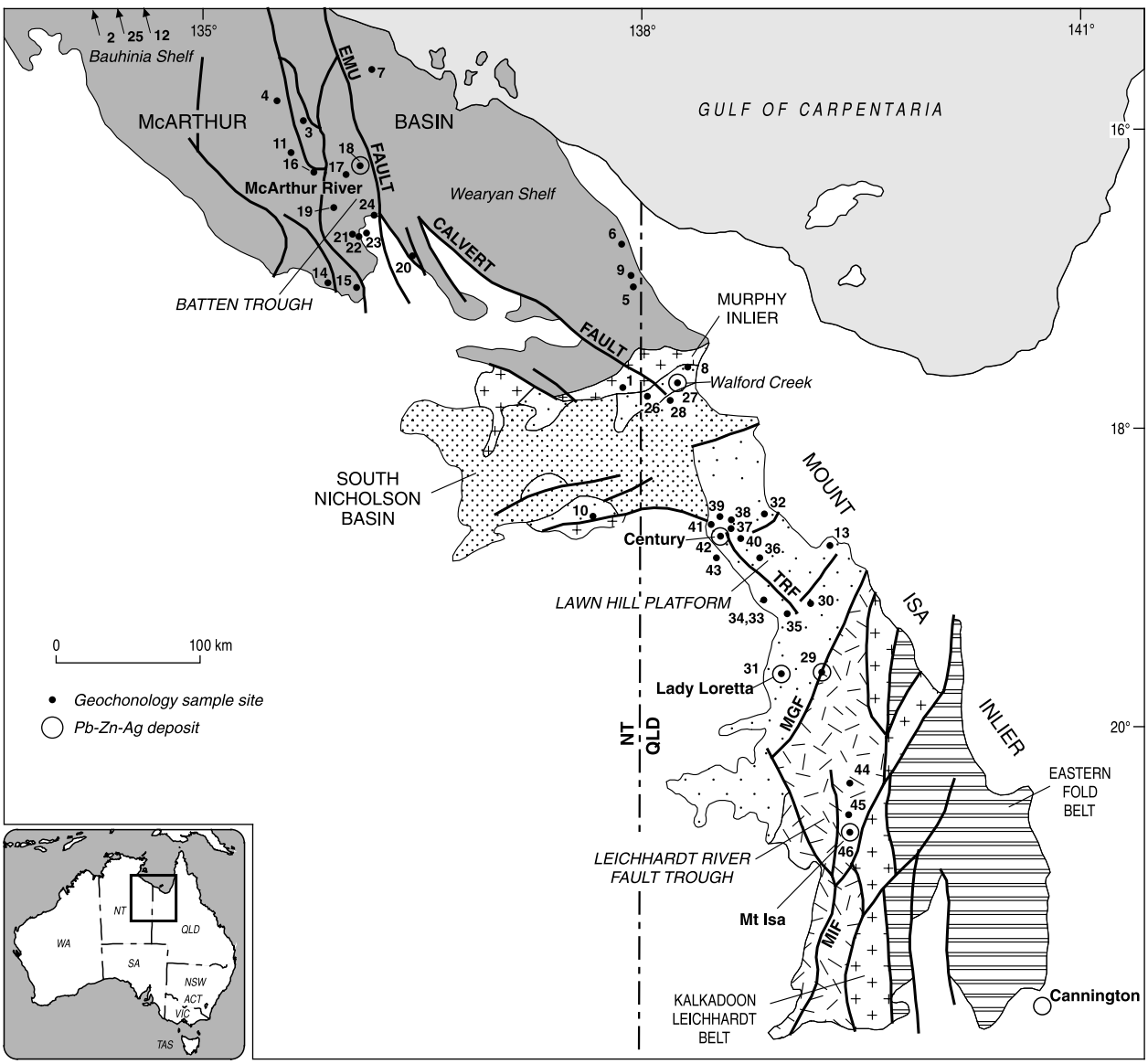


Figure 1 Locality map of the Mt Isa Inlier and part of the McArthur Basin showing the location of some of the major Pb–Zn–Ag deposits and new geochronology sites (numbered 1 to 46) listed in Table 1. TRF, Termite Range Fault; MIF, Mt Isa Fault; MGF, Mt Gordon Fault Zone.

this paper (Table 1). As well as emphasising depositional ages, the SHRIMP U–Pb zircon results include useful information on zircon provenance signatures. These detrital zircon age patterns fingerprint uplifted and eroding terrains supplying detritus to the subsiding troughs of the region. The supracrustal ages of these older eroding sequences and/or crystalline basements are moderately well controlled in northern Australia as a result of the past two decades of geochronological effort (Page 1978; Page & Sun 1998) and also through several new basement age results reported herein. An analysis of the complete OZCHRON dataset for northern Australia and implications for crustal evolution is given in Scott *et al.* (2000).

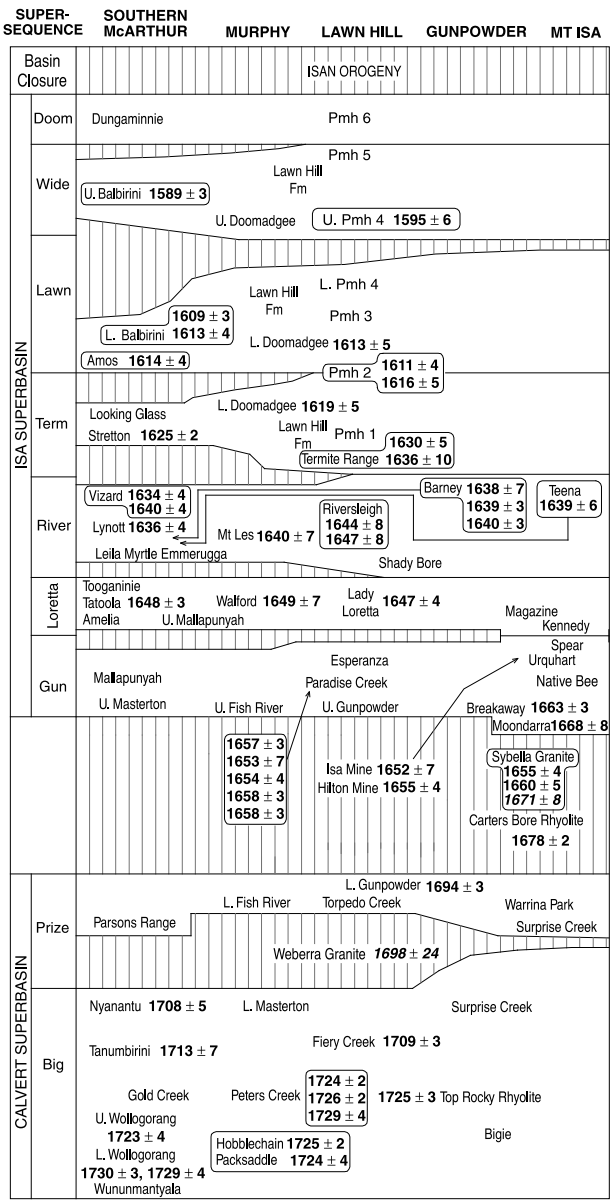


Figure 2 Chronostratigraphic event chart: summary of sequence-stratigraphic subdivisions, lithostratigraphy and SHRIMP U–Pb zircon ages for the Calvert and Isa Superbasins from the western Mt Isa Inlier, Lawn Hill Platform and southern McArthur Basin (italics: two conventional U–Pb zircon ages).

NEW SHRIMP GEOCHRONOLOGICAL RESULTS

The results of U–Pb SHRIMP dating studies carried out during the past 5 years in the NABRE project (Figure 2; Table 1) are presented here in stratigraphic order (starting with the oldest) and in regional groupings according to the lithostratigraphic unit from which the samples were taken. Details of the analytical techniques are given in Appendix 1. The use of these data and how they relate to the new and evolving sequence stratigraphic framework is then developed within this basic lithostratigraphic setting.

Basement

NICHOLSON GRANITE AND CLIFFDALE VOLCANICS

Very little U–Pb zircon dating has been undertaken in the basement rocks of the Murphy Inlier, which separates the Palaeoproterozoic succession in the McArthur Basin (to the north) from its approximate time equivalents in the Mt Isa – Lawn Hill Platform areas (to the south). The intrusive granites and volcanic rocks in the Murphy Inlier yield a variety of minimum Rb–Sr and K–Ar total-rock and mineral ages between 1820 Ma and 1770 Ma (summarised in Page 1988). Zircons from a sample (9177.9056) of a very coarse porphyritic phase of the Nicholson Granite, from the central part of the Murphy Inlier on the Seigal Sheet area, are euhedral and strongly zoned. Their high U content and resultant minor alteration has led to some Pb loss evident in 12 of the 38 analyses (Table 2*; Figure 3a). The 26 remaining analyses form a single $^{207}Pb/^{206}Pb$ cluster (χ^2 1.40) and indicate a zircon crystallisation age of 1856 ± 3 Ma. A more massive granitoid intrusion in the Nicholson Granite (9177.9057) 10.5 km to the northwest was also sampled. Zircons from this sample are generally less altered, have lower U (200–500 ppm) and normal Th/U (~0.3–0.6). Two data points (grains 135.1, 136.1) have significantly higher Th/U than the rest, and give inheritance ages of *ca* 1880 Ma and *ca* 1900 Ma, and several other data points are discordant. The pooled age derived from 50 of the 56 analyses is 1845 ± 3 Ma (χ^2 0.79; Table 3*). This igneous crystallisation age indicates that the granitoid intrusion is marginally younger than the porphyritic body, but both ages substantiate the 'Barramundi' age commonly assumed for granites in the Murphy Inlier.

A population of squat euhedral zircons from a sample of the Billicumidji Rhyolite Member of the Cliffdale Volcanics (9177.9059), collected 15 km further to the northeast, provides generally concordant SHRIMP U–Pb data (Figure 3b). A pooled age of 1851 ± 3 Ma was derived from 49 concordant points of the 51 analyses (χ^2 0.82; Table 4*) and confirms that the Nicholson Granite and Cliffdale Volcanics are coeval.

URAPUNGA FELSIC IGNEOUS ROCKS

The Mt Reid Rhyolite (previously informally referred to as Mt Reid Volcanics) and Urapunga Granite are small areas of outcropping basement in the Urapunga Fault Zone/Tectonic Ridge near Roper Bar. This west-northwest-trending basement high separates the McArthur Basin into northern and southern depocentres. A sample of the porphyritic felsic volcanic rock (9177.9024) contains zoned

euhedral zircon grains of variable Th/U; many grains have very high-U magmatic rims. A few morphologically distinct zircon cores are present but were not analysed. Some of the high-U and most of high-Th/U grains have suffered more than several per cent recent Pb loss. The concordant data from 13 remaining analyses indicate a magmatic age of 1847 ± 16 Ma (χ^2 0.60) (Table 5*; Figure 3c).

A sample of the massive undeformed granite (9177.9025) contains biotite-rich xenoliths and pegmatitic enclaves. Zircon morphology is similar to the volcanic sample, but there is a higher proportion of zircon cores. The cores are concordant and the seven core analyses have ages ranging between 2596 Ma and 1994 Ma, indicative of either complex source(s) or crustal contamination. The 19 SHRIMP analyses for the magmatic suite give a well-defined $^{207}Pb/^{206}Pb$ array (χ^2 1.15) and constrain the pooled age to 1858 ± 7 Ma (Table 6*; Figure 3c). The two results are consistent with the prediction that the intrusive and extrusive basement phases are probably coeval, and they clearly establish the 'Barramundi' age of this basement exposure.

SCRUTTON VOLCANICS

The Scrutton Volcanics unconformably underlie the Yiyintyi Sandstone in the McArthur River region and have long been considered to represent 'basement' to the sedimentary and volcanic rocks of the lower McArthur Basin succession (Plumb & Paine 1964; Pietsch *et al.* 1991). They comprise rhyolitic to dacitic lavas, ignimbrites, shallow intrusives and epiclastics. Minor, perhaps temporally unrelated, doleritic and dacitic intrusives are also present. Although they generally outcrop poorly, a sample from the central part of the Tawallah Ranges (northern Bauhinia Downs Sheet area) was dated. The SHRIMP U-Pb zircon data from 14 analyses describe a simple concordant grouping (χ^2 0.86) and give a pooled age of 1851 ± 7 Ma (Table 7*; Figure 3d). This is the magmatic age for the unit, which now can be unambiguously associated with the 'Barramundi' basement at Mt Isa (Blake 1987).

Calvert Superbasin Phase

Jackson *et al.* (2000) define the Calvert Superbasin Phase as containing those rocks aged between about 1735 and 1690 Ma that were deposited in the Mt Isa–McArthur River region, between the older Leichhardt and younger Isa Superbasins. Detailed sequence-stratigraphic studies have not been undertaken on the rocks in this superbasin, but some initial suggestions on the possible evolution for part of the succession have been presented by Jackson *et al.* (1998), and the associated U–Pb SHRIMP results are documented here.

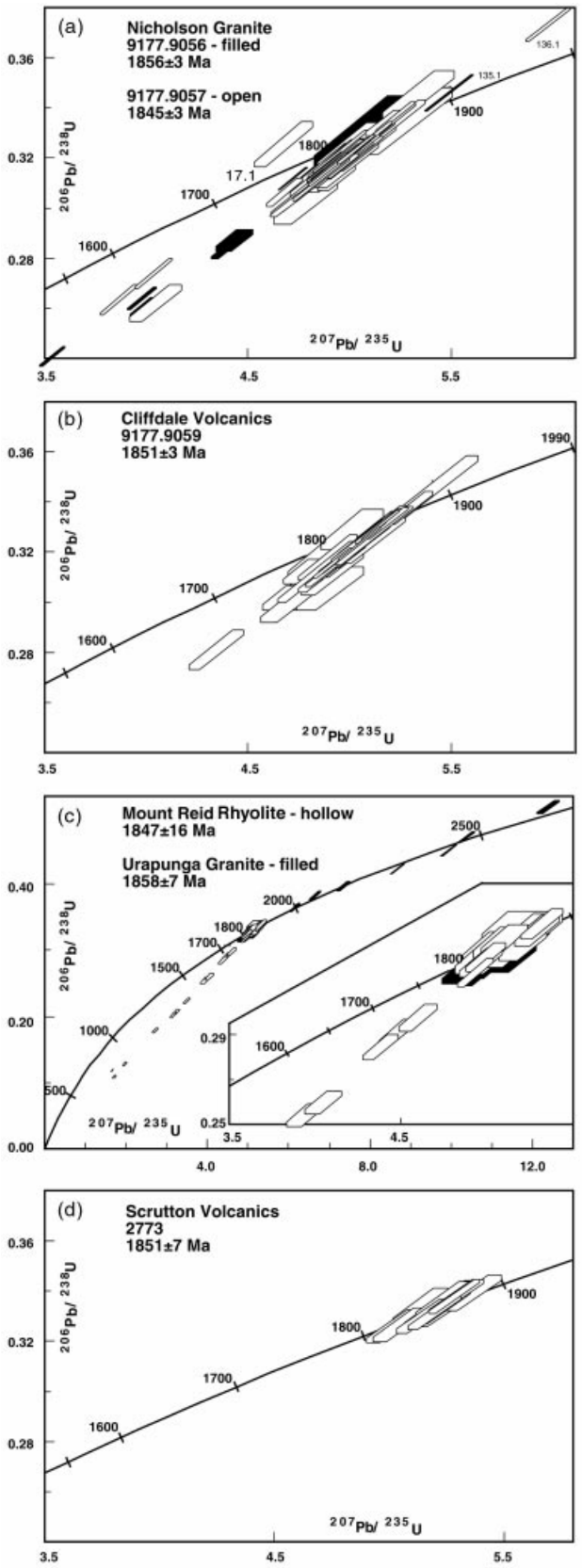


Figure 3 (a) U–Pb concordia plot of zircon SHRIMP data from two granitoids in the Nicholson Granite: porphyritic granodiorite 9177.9056, filled error boxes; massive granite 9177.9057, open error boxes. SHRIMP error boxes in this and subsequent figures are 1σ analytical uncertainties, although final pooled $^{207}Pb/^{206}Pb$ ages have errors quoted at the 95% uncertainty level. (b) U–Pb concordia plot of zircon SHRIMP data from a rhyolitic unit in Cliffdale Volcanics 9177.9059. Forty nine of the 51 analyses define the zircon crystallisation age. (c) U–Pb concordia plot of zircon SHRIMP analyses from a felsic volcanic in the Mt Reid Rhyolite (formerly Mt Reid Volcanics) (9177.9024, open error boxes) and an undeformed Urapunga Granite (9177.9025, filled error boxes). (d) U–Pb concordia plot of zircon SHRIMP data from the Scrutton Volcanics (2773) with 14 data points defining the crystallisation age.

WOLLOGORANG FORMATION

The Wollogorang Formation, comprising a lower stromatolitic dolostone, dololutite and shale succession and an upper dolomitic sandstone succession, was used as a key stratigraphic marker during the mapping of the southern half of the basin in the 1980s (Jackson *et al.* 1987; Pietsch *et al.* 1991; Haines *et al.* 1993). The type section of the Wollogorang Formation was re-examined and reassessed during the NABRE project in 1995 from a sequence-stratigraphic perspective. This revisit suggested that a mid-unit sequence boundary may separate the upper sandstone-rich part of the formation from the underlying dolostone–shale succession, and that these should be considered as parts of different basin phases and not one genetic stratigraphic unit.

Tuffaceous beds have been recognised in the Wollogorang Formation in drillholes at four widely separated localities: BMR Mt Young 2 (Figure 1, Locality 4) and BHP McA 13 (Figure 1, Locality 7) in the central part of the McArthur Basin, and CRA RC 18 (Figure 1, Locality 6) and CRA HC1, (Figure 1, Locality 5) near the southeastern margin of the basin. Tuffs from below the suspected sequence boundary were collected from all four drillholes, but only one sample was obtained from above the sequence boundary (in HC1). Preliminary results were reported in Jackson *et al.* (1998).

Mt Young 2 drillhole

The Mt Young 2 drillhole samples (7977.9066, 7977.9067) were from two 20 cm-thick beds of green claystone, 2 m apart, in about the middle of the laminated black shale facies of the formation. Jackson (1985) and Donnelly and Jackson (1988) interpreted these green claystone layers as tuffaceous beds deposited in relatively deep, low-energy sub-wave-base environments. Their vitroclastic texture is evident from fine shard-like outlines now altered to a felted mass of fine white mica and chlorite. These features are consistent with the pristine igneous morphology of the finely zoned euhedral zircon grains. The SHRIMP U–Pb data from these samples are generally concordant or sub-concordant with the exception of a few analyses. There is clear recent Pb loss and well-defined grouping in $^{207}Pb/^{206}Pb$ age (Tables 8*, 9*; Figure 4a). In sample 7977.9066, high-U grains 5 and 14 (89%, 51% Pb loss) were discarded, and the remaining 22 analyses define a $^{207}Pb/^{206}Pb$ age of 1729 ± 4 Ma (χ^2 1.21). A similarly tight $^{207}Pb/^{206}Pb$ age of 1730 ± 3 Ma (χ^2 0.54) is defined by all 25 data points from sample 7977.9067. Zircon from this sample has comparable U and lower, more uniform Th/U than in 7977.9066. The two isotopic ages are inseparable. This agreement and magmatic zircon morphologies suggest that these ages date volcanism that is contemporary with deposition in the lower part of the Wollogorang Formation.

BHP McA 13 drillhole

A sample of green 'massive' mudstone (9577.9086) was collected from 299.00–299.20 m in BHP McA 13. The sample was collected from a 5 m interval of dark grey dolomitic siltstone with prominent crystalline dolostone concretions—the 'ovoid beds' (see Donnelly & Jackson 1988). It is from essentially the same stratigraphic level as samples 9066 and 9067 in Mt Young 2. About 60 zircon grains were recovered, but the SHRIMP U–Pb data especially for the moderately high-Th/U zircons are mostly quite discordant with indicated recent Pb loss (Table 13*; Figures 5b, c). Some of the zircons have quite high Th/U and common Pb corrections. One rounded grain (509.1) has a $^{207}Pb/^{206}Pb$ age of 1835 ± 31 Ma. This and the nine analyses with high Th/U (>1.6) give scattered generally older ages and are rejected from the pooled age. The best interpreted tuff depositional age, from the remaining 10 measurements, is 1735 ± 6 Ma (χ^2 1.07). For this sample, the data are too few and mostly too discordant to rely independently on the result, but it can be considered a maximum depositional age for this part of the unit, especially as it is within error of the other ages determined for the lower Wollogorang Formation.

CRA RC 18 drillhole

A sample of green claystone was collected from the 'ovoid facies' of the Wollogorang Formation at a depth of 263.92 m in CRA drillhole RC 18. This was drilled in the central part of the Selby 1:100 000 sheet area, some 190 km southeast of McA 13. The dolostone facies and tuffaceous claystone are indistinguishable from that seen in McA 13.

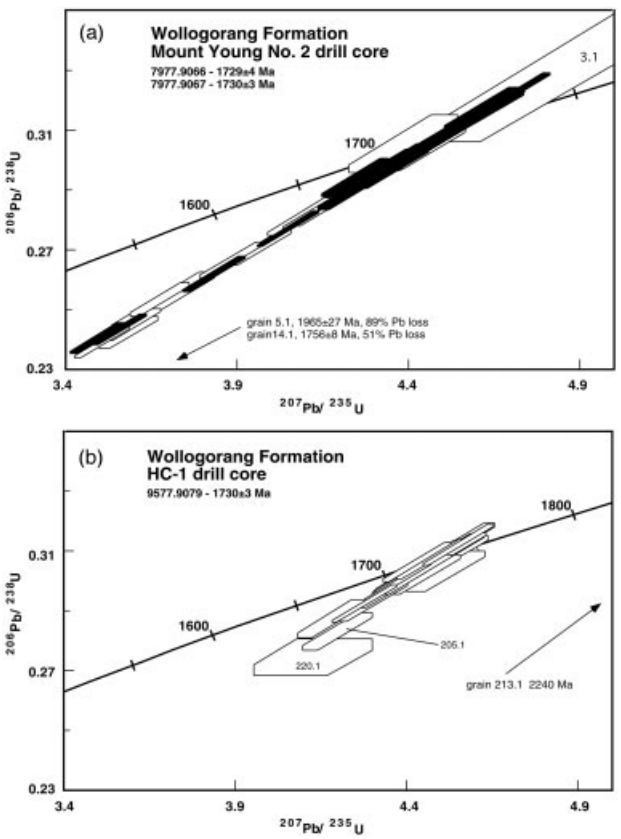


Figure 4 (a) U–Pb concordia plot of zircon SHRIMP data from the Wollogorang Formation tuff samples in Mt Young drillcore. Sample 7977.9066 (open error boxes) and 7977.9067 (darkened error boxes). (b) U–Pb concordia plot of zircon SHRIMP data from the Wollogorang Formation tuff sample in CRA drillcore HC1, 297 m, sample 9577.9079.

The sample (9577.9085) contains a sparse (only 23 grains recovered) but moderately complex zircon population. Although the zircons are euhedral and have apparently simple magmatic morphologies, some are inherited xenocrysts. Replicate SHRIMP analyses were made on many of the grains and the complex age distribution in the 41 zircon analyses is evident in Table 12* and Figures 5b, c. The population is composed of some individual grains and grain cores between 1860 and 1800 Ma, further zircon cores (commonly high Th/U) between 1800 and 1760 Ma, and a younger magmatic group (with normal Th/U) between 1750 and 1710 Ma. The group in the age range 1780–1760 Ma reflects a provenance pattern commonly observed in other sediments. The main zircon suite, interpreted as syndepositional and derived from an igneous tuff event, consists of euhedral clear grains which have a pooled $^{207}Pb/^{206}Pb$ age of 1734 ± 5 Ma (χ^2 0.68). Rejection of the more discordant data does not alter the age, which can be regarded as a maximum depositional age for the lower part of the Wollogorang Formation in this area.

CRA HC1 drillhole

CRA HC1, the only drillhole that provided samples from above and below the suspected sequence boundary, is sited approximately 60 km south of RC 18 (Figure 1, Locality 5). The lower sample (9577.9079) was collected from a 36 cm-thick, green tuffaceous claystone from a section of laminated black shale facies with incipiently developed concretions. Sample 9577.9079 yielded about 600 zircon grains, and the 25 analyses of the zoned igneous grains are given in Table 10* and Figure 4b. Some older, rounded detrital grains are also present, and one of these (grain 213) has an apparent age of 2240 Ma. The main euhedral igneous population of 22 analyses has a pooled $^{207}Pb/^{206}Pb$ age of 1730 ± 3 Ma (χ^2 1.34) when the two most discordant zircon grains (also highest Th/U) are rejected. This age is clearly in agreement with the results from the lower part of the formation from the other drillholes. A second sample (9577.9084) was collected from a 1 m-thick interval of broken green tuffaceous claystone between 234.5 and 235.5 m within a section of coarse-grained, dolomitic sandstone from the upper part of the Wollogorang Formation. Only about 20% of the sparse zircon population consists of pristine euhedral magmatic grains. The remainder is high-U altered zircons and abraded-pitted grains (of those measured from the latter, up to 2000 Ma old). Even the best igneous zircons in this tuff are somewhat altered and discordant in Pb/U, possibly because of higher U contents. The U–Pb SHRIMP data are more than 10% discordant for half of the analysed zircon grains (Table 11*; Figure 5a). From the 41 analyses, three grains (305.1 at 1997 Ma; 309.1 at 1867 Ma; and 329.2 at 1857 Ma) are clearly older than the bulk of the population. Their deletion together with all that are >90% discordant, gives a weighted mean $^{207}Pb/^{206}Pb$ age of 1725 ± 4 Ma for the 19 most concordant analyses. However, the data are not all within experimental uncertainties (χ^2 2.53). Further rejection of the oldest ages (grains 329 and 308) provides a single age distribution (χ^2 1.57) and a pooled age for the zircon population of 1722 ± 4 Ma. This age can be interpreted as a reliable maximum age for deposition. It is statistically younger than the *ca* 1730 Ma ages (at the 1% level using Student's *t*-test) reported above for the lower Wollogorang Formation, and taken at face value, suggests that there is a time break of about 8 million years between the lower and upper parts of the formation. Using the new SHRIMP information and sequence stratigraphic criteria, Jackson *et al.* (2000) assign the lower and upper parts of the Wollogorang Formation to different Associations (G and H) in the Calvert Superbasin.

Further, we note that the Hobblechain Rhyolite and other felsic extrusive rocks flanking the Murphy Inlier (e.g. Peters Creek Volcanics), which are about 200 m stratigraphically higher up in the Tawallah Group, have

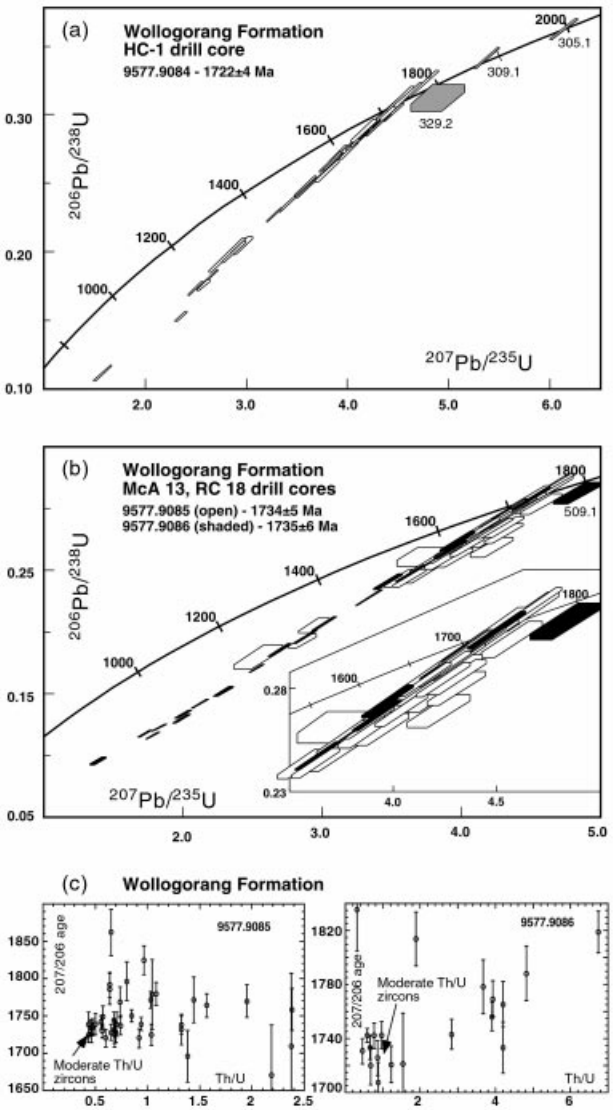


Figure 5 (a) U–Pb concordia plot of zircon SHRIMP data from the younger sandy facies of the Wollogorang Formation tuff sample in CRA drillhole HC1, 235 m, sample 9577.9084. (b) U–Pb concordia plot of zircon SHRIMP data from the Wollogorang Formation tuff samples 9577.9085 (open error boxes) and 9577.9086 (shaded error boxes). (c) Plots of Th/U *vs* $^{207}Pb/^{206}Pb$ age for zircons from Wollogorang Formation tuff samples 9577.9085 and 9577.9086. Zircons with low to moderate Th/U tend to cluster in $^{207}Pb/^{206}Pb$ age compared to the higher Th/U zircons, many of which have scattered older ages.

effectively the same age (1726 ± 2 Ma, 1724 ± 2 Ma) as the upper Wollogorang tuff. This suggests that there is essentially no time break between the two, and that the intervening Gold Creek Volcanics were deposited quite rapidly. Alternatively, these younger upper Wollogorang ages may be influenced by the presence of inherited zircons crystallised a few million years before the deposition of the tuff. Additional geochronological research is needed to assess these alternatives.

PETERS CREEK VOLCANICS

The Peters Creek Volcanics form a 10 km-wide, west-trending outcrop belt of mainly felsic igneous rocks with interstratified sedimentary and mafic volcanic rocks along the southeast margin of the Murphy Inlier. Sweet *et al.* (1981) subdivided the Peters Creek Volcanics into seven members identified by map symbols Ptp_1 to Ptp_7 and interpreted most of this igneous pile as extrusive. Later, Page and Sweet (1998) reported SHRIMP U–Pb ages of 1726 ± 2 Ma from map unit Ptp_2 (second oldest member) and 1725 ± 2 Ma from Ptp_7 (youngest member) from the western end of the outcrop belt. They interpreted this to indicate there is no discernible age difference between the two members and that 'the entire volcanic package was extruded within a few million years'. A sample (9677.9162) from the eastern end of the felsic unit mapped as Ptp_2 has also been analysed. The zircon population generally consists of zoned magmatic grains, but also includes subrounded grains, which are probably xenocrysts and were avoided in SHRIMP analyses. A pooled age of 1729 ± 4 Ma is derived from 33 data points (Table 14*; Figure 6a). However, they do not form a simple grouping (χ^2 3.19). Deletion of four discordant analyses, as well as the two oldest and two youngest data points (128.1, 131.1, 107.1, 119.1), leaves a group of 25 analyses which provide a pooled age of 1729 ± 4 Ma (χ^2 1.63). This result may suggest that felsic magmatism commenced slightly earlier than previously assumed, but is within error of the previously determined ages for the Peters Creek Volcanics. Jackson *et al.* (2000) reinterpret these 1729–1725 Ma ages as dating magmatism that is intrusive, not extrusive, in origin.

PACKSADDLE MICROGRANITE

Page and Sweet (1998) reported an age of 1725 ± 2 Ma for the Hobblechain Rhyolite, which is part of a felsic magmatic complex in the upper part of the Tawallah Group, approximately 60 km north of the Murphy Inlier. The Packsaddle Microgranite is a coarse grained felsic unit, which on field relationships appears to be comagmatic with the Hobblechain Rhyolite. A sample of microgranite was collected from close to the mapped contact between the two felsic units (Figure 1, Locality 9). It contains a zircon suite of euhedral, zoned grains that contain long apatite inclusions and tear-shaped bubbles, zircon morphological characteristics common to many felsic volcanic samples. The SHRIMP data (Table 15*; Figure 6a) define a single $^{207}Pb/^{206}Pb$ cluster (χ^2 1.17) and a weighted mean age for all 22 analyses of 1724 ± 4 Ma. There are no xenocrysts or other outliers. This igneous crystallisation age is thus strong evidence supporting a comagmatic relationship between the Packsaddle Microgranite and the Hobblechain Rhyolite.

TOP ROCKY RHYOLITE

A sample of massive rhyolite was collected from outcrop of the Top Rocky Rhyolite at the eastern end of the Carrara Range in the northwestern Lawn Hill Platform, about 90 km south of the Murphy Inlier (Figure 1, Locality 10). The Top Rocky Rhyolite is extremely weathered and, as a result, much of the magmatic zircon population (including a few subrounded detrital grains) is cracked and hematised. The SHRIMP analyses were focused only on pristine magmatic grains or remnants thereof (Table 16*). Two of the analyses (26.1, 27.1) give Cambrian ages and probably reflect detrital contaminants introduced into the rock as a result, of weathering. Some clearly older inherited grains ranging in $^{207}Pb/^{206}Pb$ age from 1933 to 1799 Ma are also present. The great majority of the zircons are, however, coherent in $^{207}Pb/^{206}Pb$ (χ^2 1.42) and the 33 data points give an age of 1725 ± 3 Ma (Table 16*; Figure 6a). The stratigraphic equivalence of the Top Rocky Rhyolite to other Calvert Superbasin igneous associations is therefore established.

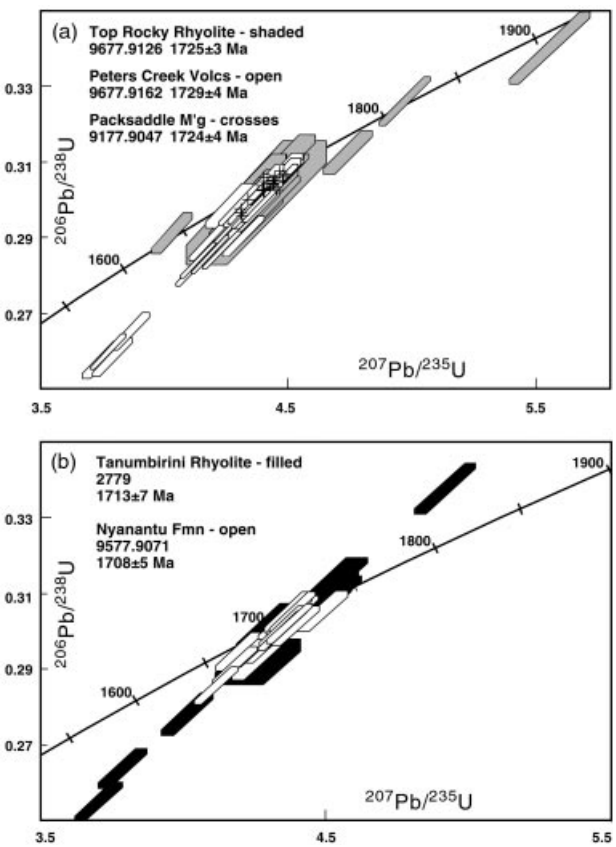


Figure 6 (a) U–Pb concordia plot of zircon SHRIMP analyses from a weathered porphyritic rhyolite in the Top Rocky Rhyolite, Carrara Range (9677.9126, shaded error boxes), rhyolite in the Peters Creek Volcanics (9677.9162, open error boxes), and a microgranite in the Packsaddle Microgranite (9177.9047, crosses). (b) U–Pb concordia plot of zircon SHRIMP analyses from a felsic volcanic in the Tanumbirini Rhyolite (2779, filled error boxes) and felsic volcanic pebbles from the Nyanantu Formation (9577.9071, open error boxes).

Table 1 Summary of new U–Pb zircon SHRIMP ages (based on *n* measurements) and locations for felsic tuffs and metasediments from various basement units and stratigraphic entities in the Isa Superbasin, Northern Territory and Queensland.

Stratigraphic unit	Supersequence	Sample no.	Standard QGNG $^{207}Pb/^{206}Pb$ age (± 95% confidence)	Easting	Northing	U–Pb Age (Ma) (± 95% confidence)	Major detrital inheritance ages (Ma)
Basement to the Leichhardt Superbasin							
Dated events in the Leichhardt period							
1. Nicholson Granite	–	9177.9056	–	807824	8031167	1856 ± 3 (n = 26)	–
1. Nicholson Granite	–	9177.9057	–	798224	8035367	1845 ± 3 (n = 50)	–
1. Cliffdale Volcanics	–	9177.9059	–	811224	8044767	1851 ± 3 (n = 49)	–
2. Mt Reid Rhyolite	–	9177.9024	–	446627	8376765	1847 ± 16 (n = 13)	–
2. Urapunga Granite	–	9177.9025	–	444327	8376065	1858 ± 7 (n = 19)	–
3. Scrutton Volcanics	–	2773	–	564325	8229767	1851 ± 7 (n = 14)	–
Dated events in the Calvert period							
8. Peters Creek Volcanics (east)	–	9677.9162	–	231321	8039571	1729 ± 4 (n = 25)	–
8. Peters Creek Volcanics (lower)[a]	–	9177.9055	–	185900	8027050	1726 ± 2 (n = 22)	–
8. Peters Creek Volcanics (upper)[a]	–	9177.9049	–	214750	8033744	1724 ± 2 (n = 24)	–
9. Packsaddle Microgranite	–	9177.9047	–	818755	8107463	1724 ± 4 (n = 22)	–
9. Hobblechain Rhyolite[a]	–	9177.9046	–	810329	8119662	1725 ± 2 (n = 17)	–
10. Top Rocky Rhyolite	–	9677.9126	–	792650	7933173	1725 ± 3 (n = 33)	–
11. Tanumbirini Rhyolite	–	2779	–	562325	8206667	1713 ± 7 (n = 22)	–
11. Nyanantu Formation	–	9577.9071	–	562225	8206667	1708 ± 5 (n = 22)	–
12. Jimbu Granite[b]	–	9477.0951	–	390327	8534865	1720 ± 7 (n = 29)	–
13. Fiery Creek Volcanics[a]	–	9420.9201	–	308040	7866230	1709 ± 3 (n = 19)	–
14. Settlement Creek Volcanics[c]	–	9577.9069	–	595226	8110967	2482 ±15 (1σ)	–
Wollogorang Formation felsic tuffs							
4. Wollogorang Formation		7977.9066	1851 ± 3 (n = 18)	555100	8245000	1729 ± 4 (n = 24)	–
4. Wollogorang Formation		7977.9067	1851 ± 3 (n = 18)	555100	8245000	1730 ± 3 (n = 25)	–
5. Wollogorang Formation		9577.9079	1851 ± 3 (n = 18)	796800	8100100	1730 ± 3 (n = 24)	–
5. Wollogorang Formation		9577.9084	1851 ± 3 (n = 18)	796800	8100100	1722 ± 4 (n = 19)	–
6. Wollogorang Formation		9577.9085	1850 ± 4 (n = 13)	795838	8151073	1734 ± 5 (n = 29)	–
7. Wollogorang Formation		9577.9086	1850 ± 4 (n = 13)	628800	8248800	1735 ± 6 (n = 10)	–
McArthur Group and Nathan Group felsic tuffs							
15. Upper Mallapunyah Formation	Loretta	9577.9041	1842 ± 7 (n = 18)	596826	8109767	<1650 ± 27	*ca* 1720, 1750, 1850–1900, 2100–2300, 2500, 3100
16. Tatoola Sandstone	Loretta	9177.9042	–	572674	8168788	1648 ± 3 (n = 49)	
17. Teena Dolomite (Coxco Member)	River	9177.9034	–	613853	8177608	1639 ± 6 (n = 25)	
18. Barney Creek Formation[a]	River	7810.9001	–	617525	8182767	1638 ± 7	
		9177.9032	–	617525	8182767	1639 ± 3 (n = 127)	1830–1860, 2740
		9177.9031	–	617525	8182767	1640 ± 3	
19. Lynott Formation (Hot Springs Mbr)	River	9577.9073	1849 ± 4 (n = 12)	592725	8163967	1636 ± 4 (n = 33)	
19. Lynott Formation (Donnegan Mbr)	River	9177.9036	–	592388	8163070	–	1760, 1820, 1860,
20. Stretton Sandstone	Term	9177.9043	–	613025	8135167	1625 ± 2 (n = 68)	1860
21. Amos Formation	Lawn	9577.9072	1849 ± 4 (n = 12)	578725	8146867	1614 ± 4 (n = 34)	
22. Lower Balbirini Dolomite	Lawn	9177.9038	1851 ± 3 (n = 17)	578799	8146934	1613 ± 4 (n = 33)	

Table 1 continued

Stratigraphic unit	Supersequence	Sample no.	Standard QGNG $^{207}Pb/^{206}Pb$ age (± 95% confidence)	Easting	Northing	U–Pb Age (Ma) (± 95% confidence)	Major detrital inheritance ages (Ma)
23. Lower Balbirini Dolomite	Lawn	9577.9074	1849 ± 4 (n = 12)	578400	8147400	1609 ± 3 (n = 44)	
24. Uppermost Balbirini Dolomite	Wide	9610.1252	1851 ± 3 (n = 14)	471900	7663200	1589 ± 3 (n = 44)	
Tuffs near Mt Birch, Roper River region							
25. St Vidgeon Formation	River	9177.9029	1851 ± 3 (n = 17)	466602	8366568	1640 ± 4 (n = 37)	
25. Nagi Formation	River	9177.9028	1851 ± 3 (n = 17)	467141	8366899	1634 ± 4 (n = 34)	1700, 1810
Fickling Group sediments and felsic tuffs							
26. Fish River Formation	Gun	9577.9078	1848 ± 7 (n = 12)	196721	8030571	<1770 Ma (n = 19)	1770, 1860, 2000, 2390
27. Walford Dolomite	Loretta	9677.9161	1849 ± 3 (n = 21)	209003	8032063	1649 ± 7 (n = 14)	1830
28. Lower Doomadgee Formation	Term	9677.9154	1849 ± 3 (n = 21)	197088	8024292	1619 ± 5 (n = 27)	1718
28. Lower Doomadgee Formation[a]	Lawn	9177.9052C	–	211721	8031071	1613 ± 5 (n = 29)	
28. Mt Les Siltstone[a]	River	9177.9051A	–	211721	8031071	1640 ± 7 (n = 19)	
McNamara Group sediments and felsic tuffs							
29. Lower Gunpowder Ck Formation	Prize 2.2	9577.9098	1854 ± 3 (n = 16)	325620	7819772	1694 ± 3 (n = 40)	
29. Paradise Creek Formation	Gun 2.3	9577.9101	1854 ± 3 (n = 16)	325920	7819872	1658 ± 3 (n = 63)	1804
29. Paradise Creek Formation	Gun 2.4	9577.9103	1851 ± 3 (n = 21)	325920	7819872	1658 ± 3 (n = 54)	1880
29. Paradise Creek Formation[a]	Gun 2.4	9277.9105	–	325821	7820172	1653 ± 7 (n = 47)	
29. Paradise Creek Formation	Gun 2.4	9577.9104	1851 ± 3 (n = 21)	325920	7819872	1654 ± 3 (n = 61)	1710–1720
30. Paradise Creek Fm @ Mellish Park	Gun 3.1	9577.9106	1851 ± 3 (n = 21)	310120	7889172	1657 ± 3 (n = 70)	
31. Lady Loretta Formation	Loretta	9577.9090	1851 ± 4 (n = 10)	295692	7811922	1647 ± 4 (n = 34)	1700–1720, 1780, 1840, 2820
31. Lady Loretta Formation	Loretta	9577.9094	1851 ± 4 (n = 10)	297327	7812064	no age determined	1770, 1970, 2450, 2700
31. Lady Loretta Formation	Loretta	9577.9096	1851 ± 4 (n = 10)	297319	7812162	*ca* 1650	1770, 1970, 2450, 2700
32. Lady Loretta Fm @ Kamarga Dome	Loretta	9710.1012	1849 ± 3 (n = 31)	270240	7921673	1648 ± 13 (n = 7)	1770, 1820, 1870, 2200
33. Riversleigh Formation	River	9677.9128 (9610.0004)	1850 ± 6 (n = 9)	269887	7889527	1647 ± 8 (n = 11)	1770, 2040
34. Riversleigh Formation	River	9677.9127 (9610.0001)	1850 ± 6 (n = 9)	269887	7889527	no age determined	1760
34. Riversleigh Formation	River	9577.9089	1849 ± 5 (n = 10)	270850	7883800	1670 max.	1670, 1770, 1855, 2475
35. Riversleigh Formation	River	9677.9133	1848 ± 4 (n = 19)	287495	7884602	<1662 ± 9 (n = 21)	1860
36. Riversleigh Formation	River	9610.0501	1848 ± 4 (n = 19)	261840	7917345	1644 ± 8 (n = 23)	1850
37. Termite Range Formation[a]	Term	9477.9103	–	247603	7930632	1630 ± 5 (n = 15)	
38. Termite Range Formation[a]	Term	9477.9104	–	259321	7919572	1636 ± 10 (n = 7)	
39. Termite Range Formation	Term	9577.9075	1848 ± 7 (n = 12)	252221	7941572	no age determined	1650, 1750, 1800–1860, 2200–2300, 2800–2900
40. Lawn Hill Formation Pmh_1		9610.0504	1848 ± 4 (n = 19)	253000	7926600	1593 ± 8 (n = 24)	1680, 1740, 1780, 1820–1890, 1960, 2100, 2500–2590
41. Lawn Hill Formation Pmh_2	Lawn	9577.9068	1848 ± 6 (n = 6)	244421	7939872	1616 ± 5 (n = 27)	1840–1850
41. Lawn Hill Formation Pmh_2	Lawn	9577.9108	1850 ± 5 (n = 10)	242796	7941130	1611 ± 4 (n = 54)	1760, 1840–1850
42. Lawn Hill Formation Pmh_4[a]	Lawn	9277.9101	–	247001	7927692	1595 ± 6 (n = 78)	1830
42. Lawn Hill Formation Pmh_4[a]	Lawn	9277.9102	–	245521	7928472	1830 ± 7 (n = 14)	1830
43. Constance Ra Ss/Lawn Hill Fm Pmh_6	Doom	9577.9077	1848 ± 7 (n = 12)	250621	7905972	1591 ± 10 (n = 27)	1822

Table 1 continued

Stratigraphic unit	Supersequence	Sample no.	Standard QGNG $^{207}Pb/^{206}Pb$ age (± 95% confidence)	Easting	Northing	U–Pb Age (Ma) (± 95% confidence)	Major detrital inheritance ages (Ma)
Mt Isa Group felsic tuffs							
44. Moondarra Siltstone	Gun	9677.9124	1849 ± 3 (n = 21)	347468	7743048	<1668 ± 8 (n = 8)	1780, 1820, 1870, 2130, 2580
44. Breakaway Shale	Gun	9677.9125	1849 ± 3 (n = 31)	345575	7743800	1663 ± 3 (n = 86)	1750, 1790, 1870
45. Urquhart Shale, Hilton	Gun	9177.9005	–	340856	7724416	1656 ± 4 (n = 28)	
45. Urquhart Shale, Hilton[a]	Gun	9177.9005	–	340856	7724416	1654 ± 5 (n = 27)	
						1655 ± 4 (n = 55, combined)	
46. Urquhart Shale, Isa[a]	Gun	7820.0001	–	341520	7707573	1652 ± 7 (n = 65)	

Depositional age results in the cover (non-basement) sequences, include a listing of pooled ages concurrently obtained from n replicate measurements of standard QGNG; the QGNG standard was not used in early stages of this work, in particular for the basement work related to the Leichhardt and Calvert periods. Numbers 1–46 refer to sites shown in Figure 1.
[a]Ages from Page & Sweet (1998).
[b]Rawlings & Page (1999).
[c]Inherited xenocryst age.

TANUMBIRINI RHYOLITE AND NYANANTU FORMATION

The Tanumbirini Rhyolite is a poorly understood massive porphyritic rhyolite that crops out in the Scrutton Ranges in the north-central Bauhinia Downs Sheet area, and in the northeast corner of the adjacent Tanumbirini Sheet area, in the southern McArthur Basin (Pietsch *et al.* 1991). Stratigraphically, it is located near the top of the Tawallah Group. In a few localities it is overlain by a polymict conglomerate and lithic sandstone called the Nyanantu Formation (Haines *et al.* 1993). This coarse volcaniclastic is more widespread in the Mt Young Sheet area, where it also seems to be stratigraphically situated at the top of the Tawallah Group.

A zoned magmatic zircon suite from a massive rhyolite (2779) collected at the southern end of the Scrutton Range, was analysed using SHRIMP. Although three of the 22 data points are somewhat discordant (Table 17*; Figure 6b) the Pb loss is recent, and the resultant pooled age of 1713 ± 7 Ma (χ^2 0.83) does not change whether or not the discordant data are included. This is interpreted as a useful stratigraphic age that dates the emplacement of the rhyolite.

A sample of arkosic debris from a pebble conglomerate (9577.9071) in the overlying Nyanantu Formation was also analysed. It yielded a population of zircons with comparably moderate U and similar Th/U (0.4–0.6) to the population from the Tanumbirini Rhyolite. There are no other morphologies or isotopic outliers from sources other than the Tanumbirini Rhyolite. The 22 SHRIMP analyses (Table 18*; Figure 6b) give a pooled age of 1708 ± 5 Ma (χ^2 1.07). This is rigorously a maximum depositional age. However, the age is consistent with its local stratigraphic relationship to the Tanumbirini Rhyolite suggesting that it is a reliable stratigraphic constraint.

Isa Superbasin phase, McArthur River region

MALLAPUNYAH FORMATION

The Mallapunyah Formation is a ~200 m-thick redbed succession of dolomitic siltstone, shale and sandstone that crops out in several faulted inliers in the southern McArthur Basin. It conformably overlies the Masterton Sandstone, the basal unit of the McArthur Group (Jackson *et al.* 1987). The Mallapunyah Formation, deposited predominantly in shallow-water to emergent environments, contains an extensive suite of evaporite pseudomorphs including large and well-developed 'cauliflower chert' nodules near the middle of the formation. Muir (1979) and Jackson *et al.* (1987) related these to deposition in extensive sabkha environments. This intermittently exposed saline–mudflat environment would, therefore, seem to be a most unusual setting within which tuffs are likely to have been preserved.

A marked gamma-ray log deflection was recorded from a thin interval of interbedded very fine-grained dolomitic sandstone, siltstone and shale from near the top of the formation, in a measured section at the southern extremity of the outcropping McArthur Basin (Figure 1, Locality 15). Anomalously high gamma-ray kicks are mostly related to

elevated K-content so sample 9577.9041, from green mudstone corresponding to the prominent gamma-ray spike, was collected to assess if it was tuffaceous. Dating of such material, if successful, would provide valuable direct age information for this lower part of the McArthur Group.

Prior to the SHRIMP analysis of the sample, sequence-stratigraphic studies suggested that this finer grained upper part of the formation might be a separate sequence from the underlying rocks. The Mallapunyah Formation was therefore tentatively subdivided into two different packages: an older, redbed evaporitic succession and a younger, finer grained clastic interval separated by a subtle sequence boundary at 200 m in the type section (Jackson *et al.* 1987 figure 53).

The presence of a heterogeneous zircon suite and the petrography of sample 9577.9041 suggests that this immature fine-grained feldspathic rock is not a tuff. SHRIMP data from 88 subrounded zircons yield a variety of ages (Table 19*; Figure 7a): 3100, 2500, 2300–2100, 1900–1850, 1800, 1750, 1730–1700, 1653 ± 9 Ma (one grain). The older detrital ages indicate a complex provenance including Tawallah Group or equivalents, Barramundi-age basement, and older Palaeoproterozoic and late Archaean terrains. Four replicate analyses on grain 350, however, produced a concordant age of 1653 ± 9 Ma (1σ; χ^2 1.18)—an imprecise maximum age for this part of the sequence. Despite this lack of precision, this new result does indicate broad correlation of the oldest part of the McArthur Group with the lower part of the McNamara Group from the Lawn Hill area. It also has important regional sequence-stratigraphic implications as it indicates that the upper part of the Mallapunyah Formation must be younger than 1653 ± 27 Ma. Within this error range, it could conceivably fit within the Gun Supersequence (range 1670–1655 Ma) or within the succeeding Loretta Supersequence (range 1655–1645 Ma). These two supersequences are separated by a sequence boundary, at about 1655 Ma, that has been identified on seismic and in outcrop in the Lawn Hill area near the base of the Lady Loretta Formation (Bradshaw & Scott 1999). Jackson *et al.* (2000c) discuss their reasoning behind allocating the lower Mallapunyah Formation to the Gun Supersequence and the upper part of the Mallapunyah Formation to the Loretta Supersequence (as shown on Figure 2).

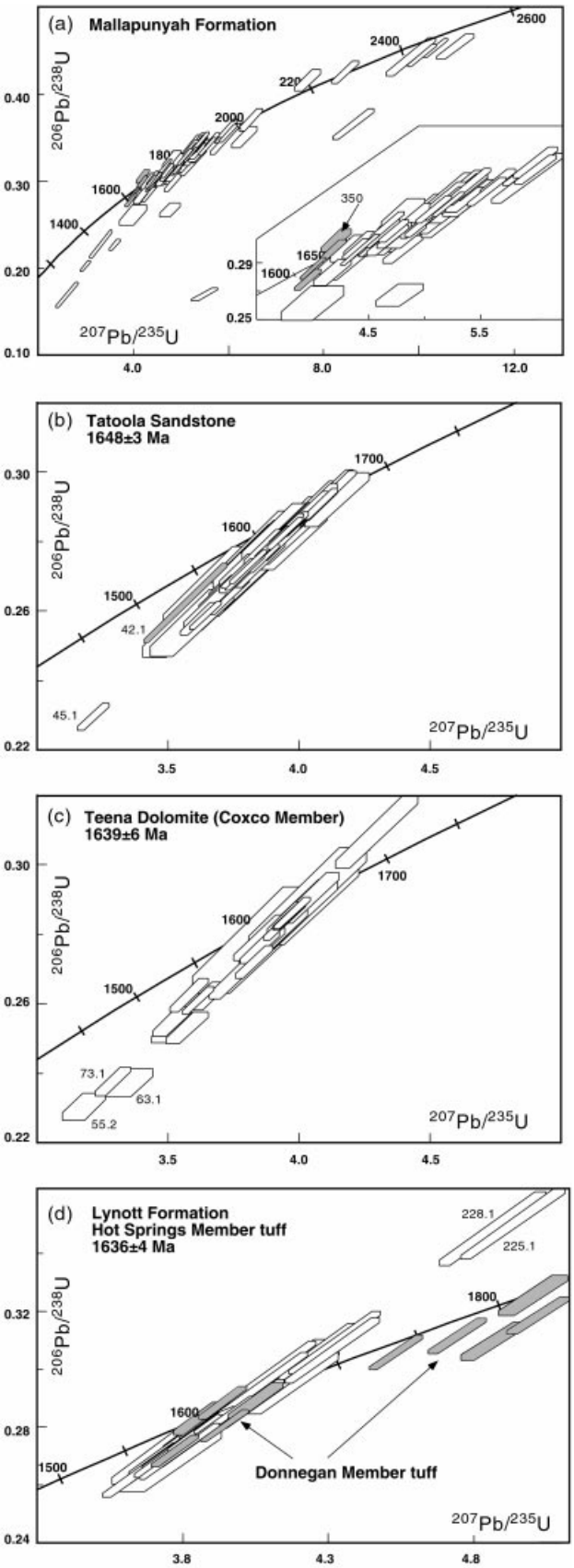


Figure 7 (a) U–Pb concordia plot of zircon SHRIMP data from green shale in the Mallapunyah Formation sample 9577.9041. Replicate analyses on the youngest detrital grain (350) are shaded grey, indicating a maximum depositional age of about 1650 Ma. (b) U–Pb concordia plot of zircon SHRIMP data from a tuffaceous unit in Tatoola Sandstone sample 9177.9042. All 49 analyses, including the youngest analysis 42.1 (shaded grey), define the zircon crystallisation age. (c) U–Pb concordia plot of 25 zircon SHRIMP analyses from a pink tuff (9177.9034) in Coxco Member at the top of the Teena Dolomite. Exclusion of the three most discordant data points does not affect the 1639 ± 6 Ma pooled zircon crystallisation age. (d) U–Pb concordia plot of zircon SHRIMP data from the Hot Springs Member (9577.9073, 35 analyses, open error boxes) and Donnegan Member (9177.9036, shaded error boxes) of the Lynott Formation. Ten of the 11 analyses from the latter show a variety of ages indicating its reworked nature; discordant analysis 203.1 not shown.

TATOOLA SANDSTONE

The Tatoola Sandstone is a distinctive white-weathering quartz sandstone that separates the predominantly stromatolitic dolostone units of the Amelia Dolomite and Tooganinie Formation in the lower part of the McArthur Basin succession. Stratigraphically, it is located about 200 m above the upper Mallapunyah Formation described above. Because Jackson *et al.* (1987) described this lower part of the basin succession as conformable, the Tatoola Sandstone would be expected to be only slightly younger than the Mallapunyah Formation. The Tatoola Sandstone is commonly around 100 m thick and crops out throughout a similar area of the basin as the Mallapunyah Formation. In most areas it consists of wavy-bedded, flaggy, fine-grained quartz sandstone. Jackson *et al.* (1987) interpreted the unit as rapidly deposited in storm- and wave-dominated environments, hardly the sort of setting where tuffs are likely to be preserved, due to the extensive reworking of the depositional interface.

A resistant, thicker, dark siliceous bed near the base of the outcropping Tatoola Sandstone at Leila Creek (Figure 1, locality 16) was sampled as a possible tuffaceous interval (sample 9177.9042). This bed contains lenticular lithic fragments and quartz and feldspar crystals in a fine groundmass. The zircon population is a pristine magmatic group consisting of clear, squat, euhedral, mainly double-terminated grains. The U–Pb data (Table 20*; Figure 7b) from all 49 SHRIMP analyses describe a normal distribution in $^{207}Pb/^{206}Pb$ (χ^2 1.11). Duplicate measurements within grains 9 and 28 are in good agreement. The data are concordant to subconcordant and give a pooled $^{207}Pb/^{206}Pb$ age of 1648 ± 3 Ma. Although grain 45.1 has 19% Pb loss, its deletion makes no difference to the age, which can be interpreted as a firm stratigraphic age for this unit. This age is indistinguishable from the maximum age reported above for the upper Mallapunyah Formation. Southgate *et al.* (2000) have used this new SHRIMP age information, together with detailed gamma logs of the measured sections, to associate the upper part of the Mallapunyah Formation, the Amelia Dolomite, Tatoola Sandstone and Tooganinie Formation as one continuous depositional episode called the Loretta Supersequence (Figure 2). In regional lithostratigraphic terms, this succession is the approximate time equivalent of the Lady Loretta Formation on the Lawn Hill Platform, and the Kennedy and Magazine Siltstones near Mt Isa.

TEENA DOLOMITE

The Teena Dolomite is a thin dolostone unit underlying the Barney Creek Formation, the host to the McArthur River Zn–Pb–Ag deposit. The Teena Dolomite is generally around 100 m thick and has been subdivided into two members, a lower unnamed member and the upper Coxco Dolomite Member. The latter is distinguished in outcrop and drillholes by the presence of numerous, pink, buff or orange-weathering K-rich mudstones and ubiquitous, radiating, aragonite crystal pseudomorphs colloquially called 'Coxco Needles'. The palaeogeographic setting of the unit is varied. Recent work by Winefield (2000) indicates that its evolution is more complex than that given in Jackson *et al.* (1987) who favoured deposition in a hypersaline lacustrine environment. The Teena Dolomite marks a major lithostratigraphic change in the McArthur Group where laterally uniform, peritidal–shallow shelf, stromatolitic carbonates are replaced by interstratified fine-grained dolomitic clastics and breccia beds of the Barney Creek Formation, which, in places, were deposited adjacent to syndepositional, strike-slip, growth faults. On the apparent polar wander path for the Palaeoproterozoic (Idnurm 2000) this unit sits on one of the major hairpin bends. Loutit *et al.* (1994) and Southgate *et al.* (2000) interpret these bends as due to major interplate reorganisation. Direct dating of this unit would therefore help to constrain the character of this important tectonic event.

The siliceous pink beds in the Coxco Dolomite Member vary from a few millimetres up to many centimetres in thickness. Some of these 'pink beds' are uniform in thickness, concordant with bedding and occur in sections of laminated (quiet water) carbonates and are interpreted as probable tuffs. However, petrologically similar K-rich material also occurs in very irregular 'beds' or veins that commonly cross-cut the stratification. As the formation contains evaporite relicts (Jackson *et al.* 1987; Winefield 2000) and is located stratigraphically near the base-metal mineralisation (an area affected by several episodes of fluid migration), it is likely that some of these unusual K-rich beds are not genuine air-fall tuffs. Davidson (1998) described several complex alkali alteration products at this stratigraphic level in the basin which have produced 'tuffaceous looking' beds.

The tuff sample analysed (9177.9034) was collected from a series of low outcrops on the east side of the McArthur River, 6 km southwest of the McArthur Mine. On the Bauhinia Downs map (Pietsch *et al.* 1991) these outcrops are shown as lower Teena Dolomite, but because of the common presence of pink beds it is likely the sample actually comes from the upper part of the Teena Dolomite, the Coxco Dolomite Member.

The 90 cm-thick massive pink bed (9177.9034) is the uppermost of three such beds that are preserved concordantly in parallel-laminated dolomitic siltstones. Only 38 zircon grains were extracted from this sample. They included a minority of high-U, altered metamict grains that could not be satisfactorily analysed, as experience suggests that these would be inherited material much older than the majority of the zircon suite. Twenty-five SHRIMP analyses (Table 21*; Figure 7c) from the remaining 22 clear squat euhedral zircon grains yield a coherent cluster in $^{207}Pb/^{206}Pb$ and an age of 1639 ± 6 Ma (χ^2 0.78). Exclusion of three discordant data points (55.2, 73.1, 63.1) does not affect this age determination.

Rigorously, this zircon crystallisation age of 1639 ± 6 Ma is a maximum depositional age for the unit, and is inseparable from the age of the immediately overlying Barney Creek Formation tuffs at 1639 ± 3 Ma, 1638 ± 7 Ma, and 1640 ± 3 Ma (Page & Sweet 1998). This indicates that these two units were deposited without any significant time break between them. This 1639 ± 6 age is, however, about 9 million years younger than the age described above from the Tatoola Sandstone (1648 ± 3 Ma) which sits about 1200 m stratigraphically below the Teena Dolomite. Jackson and Southgate (2000) use these new age data, gamma-ray curves from composite sections, facies changes and the apparent

polar wander path results to suggest a major break between the two units and thereby separating the Loretta and River Supersequences (Figure 2).

LYNOTT FORMATION

Jackson *et al.* (1987) defined the Lynott Formation as the basal unit of the ~1000 m-thick Batten Subgroup, the upper subdivision of the McArthur Group. They describe it as overlying the older Umbolooga Subgroup with local unconformity. The Reward Dolomite at the top of the Umbolooga Subgroup has not been dated, and therefore the 1640 Ma ages from the Barney Creek Formation, described above, are the youngest age yet obtained from the Umbolooga Subgroup. These Barney Creek Formation tuffs are located about 1000 m below the top of the Reward Dolomite. Two samples from the Lynott Formation were dated. These were collected from near Leila Hill (35 km southwest of McArthur mine) where the formation is about 500 m thick. The formation comprises three distinct members: a basal, largely deeper water, dolomitic siltstone unit called the Caranbirini Member; a middle cherty stromatolitic unit called the Hot Springs Member; and an upper evaporitic redbed unit called the Donnegan Member. Overall these units form a shallowing-upwards succession. Sample 9577.9073 is from near the top of the Hot Springs Member and 9177.9036 is from near the top of the Donnegan Member.

Sample 9577.9073 is a 1 cm-thick pink bed located in a section of flat-laminated, fine-grained dolarenite and dolomitic siltstone. The dolarenite contains load structures, intraclasts and climbing- ripples-in-drift indicating a high-energy environment with a high sedimentation rate. The interbedded dolomitic siltstone is parallel-laminated and was probably deposited in much lower energy settings. The sampled bed has a sharp planar base, contains ripples-in-drift and a scoured top; reworked clasts have been incorporated into the overlying dolostone bed. The sampled bed was interpreted as a reworked, waterlain, K-rich tuffaceous mudstone.

Petrographically, sample 9577.9073 is a tuff, as it has a more-or-less bimodal size distribution of fine, wispy quartzo-feldspathic matrix supporting larger quartz and feldspar crystals. Most of the zircon consists of well-faceted euhedral grains, although some of these are cracked and filled with alteration products; other rounded, clearly detrital zircons were avoided in the SHRIMP analysis (Table 22*; Figure 7d). The zircon suite has low U and moderately high Th/U (0.9–1.1), which seems to be characteristic of many of the tuff zircon suites. It is not obvious why two (grains 225.1, 228.1) of the 35 analyses are reversely concordant. The rest of the data is concordant and the 33 analyses define a single cluster in $^{207}Pb/^{206}Pb$ (χ^2 0.92) and an age of 1636 ± 4 Ma. This result can be confidently interpreted as a depositional age for this part of the Lynott Formation. This weighted mean age is only slightly younger (2 to 3 million years) than the ages obtained from the Barney Creek Formation, which geologically is quite surprising. It indicates that—(i) the deposition of the upper Barney Creek Formation; (ii) the deposition of the overlying Reward Dolomite; (iii) the structural event that produced the local unconformities between the two subgroups; and (iv) the deposition of the Caranbirini and lower part of the Hot Springs Members—all took place rapidly. Limited sequence-stratigraphic work has been undertaken on this part of the succession in the McArthur River region. However, using this together with constraints from the more extensively studied Lawn Hill region, Southgate *et al.* (2000) now define the middle part of the McArthur Basin succession (the package from the Leila Sandstone up to and including the Yalco Formation) as the River Supersequence. One important consequence of this is that the previously defined break between the Umbolooga and Batten Subgroups, which is situated at the top of Reward Dolomite/base of Lynott Formation, is now seen as a much less significant stratigraphic surface.

An attempt to date a similar looking felsic tuff band in the younger Donnegan Member (sample 9177.9036) was not successful in terms of defining a depositional age. At this location the pink beds are interbedded with chocolate-coloured siltstone, ripple-bedded silty dolostone, and 'cauliflower' chert nodules (after nodular sulfate evaporite precursors). The 2 cm-thick pink bed that was sampled has a knife-sharp base and a wavy, eroded top surface. However, the tuffaceous horizon contained few pristine igneous zircons and was dominated by subrounded abraded zircons, many of which are altered and most of which are reworked/detrital in origin. The SHRIMP results on 11 of these grains (Table 23*; Figure 7d) are too few to interpret. However, they include five detrital zircons at about 1800–1870 Ma and 1760 Ma, and a few grains whose U–Pb ages overlap and are consistent with the 1636 ± 4 Ma age of the Hot Springs Member.

ADDITIONAL McARTHUR BASIN SHRIMP AGES

Results of studies of five additional samples from the upper part of the McArthur Basin succession (*ca* 1625–1580 Ma) are presented in Jackson *et al.* (2000b). They use these new SHRIMP data, in conjunction with sequence-stratigraphic results, to subdivide the upper part of the Batten Subgroup and the overlying Nathan Group into four supersequences—the Term, Lawn, Wide and Doom Supersequences. In the McArthur area, these supersequences appear to be separated by significant gaps in the rock record (Figure 2). The break between the McArthur and Nathan Groups (at the base of the Amos Formation) is shown to be a minor stratigraphic discontinuity involving little loss of geological section. Two major surfaces with significant loss of section are present within the Balbirini Dolomite in what was previously interpreted as a conformable and continuously deposited succession.

All of the McArthur Group SHRIMP ages are summarised in Figure 8. The veracity of these depositional ages is substantiated by the predictable gradual younging in mean ages (amount of offset of the peak to the left of the base line at 1650 Ma) from the oldest unit (Tatoola Sandstone) to the youngest unit (uppermost Balbirini Dolomite). With these new data the various McArthur Group formations can be more confidently related to Palaeoproterozoic supersequences in the Lawn Hill – Mt Isa area (Southgate *et al.* 2000).

Isa Superbasin phase, Roper River region

Samples of pink beds in a dolomitic succession near Mt Birch (~230 km northwest of the McArthur mine) were analysed. Dunn (1963) informally defined the succession at Mt Birch as 'Vidgeon Formation' and suggested equivalence to some part of the McArthur Group succession. Based on recent Northern Territory Geological Survey – Australian Geological Survey Organisation mapping, the succession at Mt Birch is now divided into two formations: a lower unit, the St Vidgeon Formation consisting predominantly of stromatolitic carbonates, and an upper unit, the Nagi Formation consisting largely of fine-grained clastics (Abbott *et al.* 2000). The base of the Nagi Formation is marked by an obvious erosion surface overlain by white-weathering, cliff-forming, coarse-grained quartz sandstone. A number of pink cherty beds exposed above and below this sandstone have several petrographic tuffaceous-like features including possible relict shards and a bimodal grainsize distribution with angular quartz and feldspar phenocrysts. Sample 9177.9029 is located about 5 m stratigraphically below the base of the Nagi Formation, and sample 9177.9028 is located approximately 30 m above it. Both contain pristine euhedral zircons that have preservation of small delicate crystal facets; there are no signs of mechanical abrasion in sample 9177.9029. The SHRIMP U–Pb data (Table 24*; Figure 9) for zircons from the older St Vidgeon Formation tuff (9177.9029) provide an age of 1640 ± 4 Ma based on all 37 generally concordant data points closely grouped in $^{207}Pb/^{206}Pb$ (χ^2 1.04). Although one grain (37.1) is somewhat discordant, its deletion makes no difference to the calculated age.

Sample 9177.9028, which overlies the white sandstone unit, contains a few inherited, although not obviously abraded zircons. These three grains (133, 128, 134) have higher U than the main suite, have undergone ~50% Pb loss, and have minimum $^{207}Pb/^{206}Pb$ ages of 1808 Ma, 1699 Ma, and 1687 Ma (Table 25*). In the main magmatic suite, apart

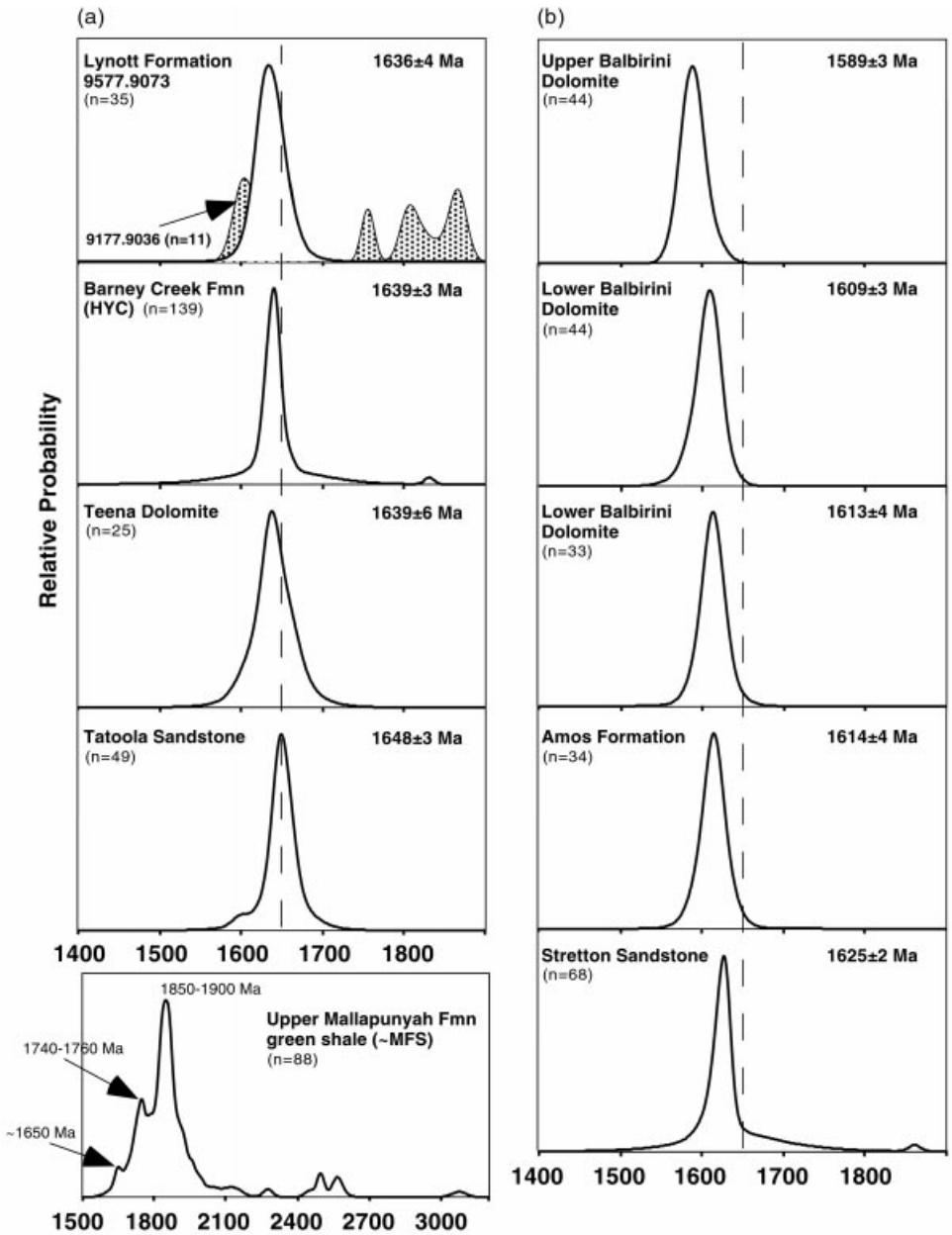


Figure 8 (a) Probability distribution plots of SHRIMP zircon ages from various tuffaceous units and shaly sediments (ascending stratigraphic order) in the lower McArthur Group, showing total number of SHRIMP analyses (n) and pooled zircon $^{207}Pb/^{206}Pb$ (depositional) ages. Dashed vertical line is 1650 Ma reference. (b) Probability distribution plots of SHRIMP zircon ages from various tuffaceous units (ascending stratigraphic order) in the upper McArthur Group and Nathan Group (Jackson *et al.* (2000b), showing total number of SHRIMP analyses (n) and pooled zircon $^{207}Pb/^{206}Pb$ (depositional) ages. Dashed vertical line is 1650 Ma reference.

from grain 107.1 which has very high Th/U (4.4) and is also discordant, the remaining 34 analyses consist of typically high-Th/U zircons, and are concordant or up to 13% discordant. As they have a single $^{207}Pb/^{206}Pb$ pattern (χ^2 0.90), the resultant age of 1634 ± 4 Ma can be unambiguously interpreted as the zircon igneous crystallisation age for the Nagi Formation tuff. Rejection of the five data points that have 10–13% recent Pb loss barely alters the pooled age.

Although the two ages cannot be separated statistically, the apparent age difference may well be real as the SHRIMP data were all collected in the same analytical session. Geologically, there is a marked basinward shift in facies at the base of the Nagi Formation, following a period of erosion, so a break in the order of a few million years would not be unrealistic. Previous tentative correlations of the stromatolitic carbonates and laminated dolomitic siltstones in the upper part of the St Vidgeon Formation with the Teena Dolomite and Barney Creek Formation (respectively) are supported by the 1640 ± 4 Ma age from the lower sample (St Vidgeon Formation). The younger Nagi Formation is lithologically very similar to the Stretton Sandstone in the McArthur Basin and on purely lithostratigraphic criteria the two would almost certainly be correlated. However, its 1634 ± 4 Ma age indicates that the Nagi Formation is the lateral equivalent of the Lynott Formation (1636 ± 4 Ma) and not the younger Stretton Sandstone (1625 ± 2 Ma). The coarser facies at Mt Birch probably indicates that the nearby Urapunga Tectonic Ridge was an active source of quartz detritus at this time (as well as much earlier). In sequence-stratigraphic terms both the St Vidgeon and Nagi Formations fall within the River Superseqence of Southgate *et al.* (2000).

Isa Superbasin phase, southern flank of the Murphy Inlier

The Fickling Group is a mixed sedimentary succession exposed in a narrow belt along the northern margin of the Lawn Hill Platform where units of the Isa Superbasin onlap the Murphy Inlier. The Fickling Group is at least 950 m thick (Sweet *et al.* 1981) and is divided into four formations: the Fish River Formation, Walford Dolomite, Mt Les Siltstone and Doomadgee Formation. Existing geochronological constraints for the Fickling Group come from SHRIMP dating of tuffs in the Mt Les Siltstone (1640 ± 7 Ma) and upper Doomadgee Formation (1613 ± 5 Ma) in the vicinity of the Walford Creek Pb–Zn deposit (Page & Sweet 1998). These results provided good chronostratigraphic links to the McArthur Basin and southern Lawn Hill Platform sedimentary successions. This paper provides further geochronological constraints for the Fickling Group, dating its oldest and youngest components.

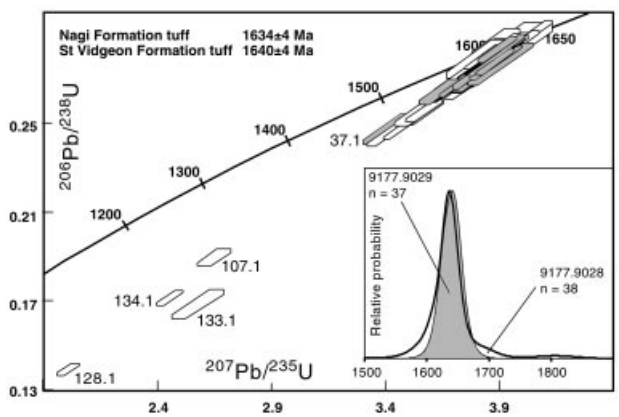


Figure 9 U–Pb concordia plot and probability distribution plots of zircon SHRIMP data from cherty pink tuffites in the St Vidgeon and Nagi Formations. Shaded data boxes are 37 analyses from a St Vidgeon Formation tuffite (9177.9029) below the white sandstone marker. Open error boxes are 38 analyses from a Nagi Formation tuffite (9177.9028) above the white sandstone marker. Numbered discordant analyses are referred to in the text.

FISH RIVER FORMATION

The Fish River Formation is a ~250 m thick, sandstone-dominated succession deposited above a major unconformity that incises into the Peters Creek Volcanics. The formation comprises three mapped members: a basal conglomeratic sandstone with rare stromatolitic carbonate beds (Pff1) up to 150 m thick at Wire Creek; a middle, thinly interbedded sandstone–siltstone–shale member (Pff2) ~60 m thick at Wire Creek; and an upper very coarse-grained sandstone member (Pff3), up to 100 m thick (Sweet *et al.* 1981). Based on broad sequence-stratigraphic correlations with the northern Lawn Hill Platform, Bradshaw *et al.* (2000) associate the lower two members with the Prize Supersequence and the upper member with the Gun Supersequence (Figure 2).

Thin bedded green (?tuffaceous) siltstone–shale in the Pff2 member of the Fish River Formation (200 m in the NABRE Wire Creek section: Southgate *et al.* 1999) was sampled in 1995. The sample (9577.9078) yielded a suite of morphologically complex, abraded and rounded zircons. These are unlikely to be of tuffaceous origin and are probably entirely reworked detritus. There is limited scope for interpreting the SHRIMP data (Table 26*; Figure 10a) because there are only 19 data points, and also because most of them are discordant. Nevertheless, individual minimum $^{207}Pb/^{206}Pb$ ages define four detrital groups: a few grains at 2390, 2000 and 1770 Ma, with the majority having a Barramundi-age signature of around 1860 Ma. The youngest few detrital grains suggest a maximum depositional age of about 1770 Ma for the Fish River Formation. This is clearly a poor constraint as improved chronostratigraphic control is provided by its relationship to the underlying *ca* 1720 Ma Peters Creek Volcanics and the overlying Walford Dolomite (below).

WALFORD DOLOMITE

The Walford Dolomite comprises oolitic, stromatolitic, and intraclastic dolostone with a maximum thickness of 400 m along the southern flanks of the Murphy Inlier (Sweet *et al.* 1981). The base of the formation is sharp and was originally interpreted as a conformable contact by Sweet *et al.* (1981). However, NABRE-measured sections document erosion on this surface and suggest that it is a regional unconformity (Lindsay & Wells 1997).

A sample was collected for SHRIMP dating from a 10 cm-thick pink ?tuffaceous interval, 18 m above the base of the

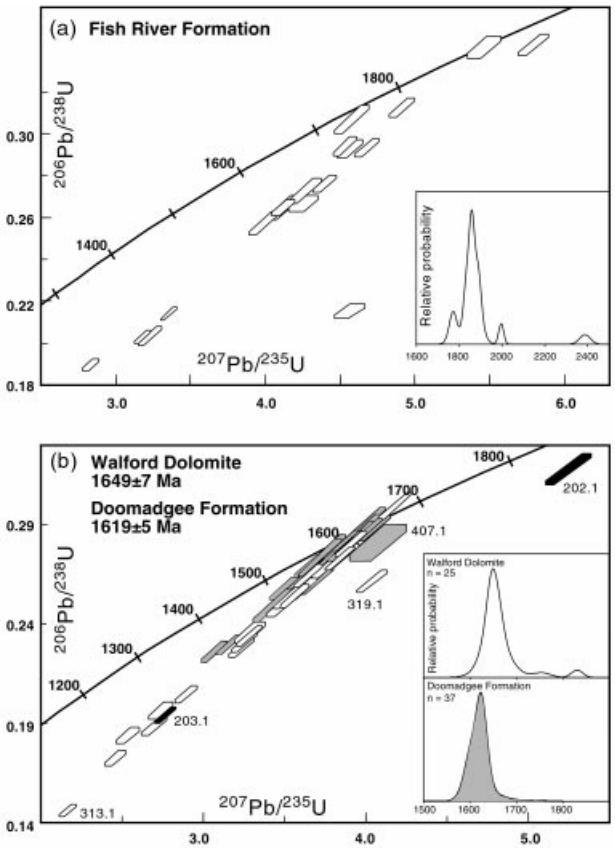


Figure 10 (a) U–Pb concordia and probability distribution plots of zircon SHRIMP data from a green shale in the Fish River Formation, lower Fickling Group (9577.9078) indicating a variety of detrital ages and a Barramundi-age peak at about 1860 Ma. (b) U–Pb concordia plot and probability distribution plots of zircon SHRIMP data from the middle and upper Fickling Group. Open error boxes represent 25 zircon analyses (including replicates) from a tuffaceous sediment in the Walford Dolomite (9677.9161). Black error boxes for two zircon analyses from a second sediment in the Walford Dolomite (9677.9160). Shaded data boxes are 37 zircon analyses from a tuff in the Doomadgee Formation (9677.9154).

Walford Dolomite, in a creek section 13 km due east of the type section. The sample (9677.9161) contains a sparse population (38 grains) of generally euhedral zircons together with a few rounded grains. Some zircons are high Th/U and are variably altered. Twenty-five SHRIMP analyses, including several intra-grain replicates, on 16 grains were obtained (Table 27*). There are two inherited (discordant) data points with $^{207}Pb/^{206}Pb$ ages of 1831 and 1755 Ma.

The zircon suite has a relatively narrow range in U, but very variable and high Th/U. Thus, although the other 23 analyses define a single $^{207}Pb/^{206}Pb$ population (χ^2 0.82) and an apparent age of 1652 ± 6 Ma, most of the high-Th/U grains (Th/U 2–5) have suffered extensive recent Pb loss, more or less proportional to Th content (Table 27*; Figure 10b). It is preferable to assess the tuff zircon age from only the concordant and subconcordant data, and inclusion of only those zircon analyses that are >85% (n = 14) or >90% (n = 10) concordant gives a pooled $^{207}Pb/^{206}Pb$ age of 1649 ± 7 Ma (χ^2 0.79). This is the zircon crystallisation age, and its stratigraphic consistency with Page and Sweet's (1998) 1640 ± 7 Ma age for the overlying Mt Les Siltstone suggests that 1649 ± 7 Ma is a reasonable depositional age for this Walford Dolomite tuff.

A second tuffaceous mudstone from the Walford Dolomite (9677.9160: Table 27*) contained only a few rounded zircon grains. Only two SHRIMP $^{207}Pb/^{206}Pb$ ages of 1952 and 1685 Ma were measured.

On the Amoco seismic grid, the Isa Supersequence onlaps to the north so that lower units were not deposited in this area (Bradshaw & Scott 1999) and the Walford Dolomite represents the upper part of the Loretta Supersequence. The Walford Dolomite thus correlates to the Lady Loretta Formation of the McNamara Group in the south (Southgate *et al.* 2000).

MT LES SILTSTONE

The Mt Les Siltstone is divided into five mappable members (Pfl1–Pfl5) and varies from 55 to 165 m in thickness. The formation consists largely of interbedded fine-grained sandstone, siltstone and shale with rare dolostone beds. The base of the formation is a sharp surface that is now interpreted as a regional unconformity marking the River Supersequence boundary (Krassay *et al.* 2000a). The bulk of the formation comprises finely laminated, black, carbonaceous shale, silty shale and siltstone. Regional peaks in gamma-ray readings commonly correspond to the finest grained and most carbonaceous parts of the formation. A tuffaceous sample from the Mt Les Siltstone at the Walford Creek Zn–Pb–Cu–Ag deposit has been previously dated as 1640 ± 7 Ma (Page & Sweet 1998). Regionally, the Mt Les Siltstone correlates with the base-metal-rich, middle to upper parts of the Riversleigh Siltstone on the central Lawn Hill Platform. Both form part of the River Supersequence.

DOOMADGEE FORMATION

The Doomadgee Formation, comprising thinly bedded shale, siltstone and sandstone with minor conglomerate and dolostone, is divided into three mappable members, Pfd1, Pfd2 and Pfd3 (Sweet & Slater 1975). The base of the formation is a sharp, erosional contact overlain by conglomeratic sandstone in the west and dolarenite in the east. Sequence-stratigraphic correlations show that the basal formation boundary has been mapped in two different stratigraphic positions across the Murphy Inlier. At Wire Creek and in drillholes along the western part of the outcrop belt the base of the Doomadgee Formation is interpreted at an unconformity beneath conglomeratic sandstone within the River Supersequence (Bradshaw *et al.* 2000; Krassay *et al.* 2000a). However, to the east near the Walford Prospect (Rohrlach *et al.* 1998) the base of the Doomadgee Formation is placed at the base of a coarse-grained dolarenite within the Term Supersequence.

A previously analysed tuff from the lower Doomadgee Formation yielded a depositional age of 1613 ± 5 Ma (Page & Sweet 1998). Another tuff in the lower Doomadgee Formation was sampled from drillcore in Pasminco drillhole GC 57 at 164.7 m. The sample, from a 50 cm-thick, green clay layer within a section of grey-green interbedded siltstone and shale, contains convincing relict shard-like

textures and relatively homogeneous, squat, euhedrally zoned igneous zircons. Apart from one aberrantly old zircon analysis (407.1: 1718 Ma), the other 36 zircon grains have high Th/U, are all coherent in $^{207}Pb/^{206}Pb$ age (χ^2 1.33) and have a weighted mean of 1619 ± 4 Ma (Table 28*; Figure 10b). If the nine analyses that are more than 8% discordant are discarded, the remaining 27 (χ^2 1.22) define an age of 1619 ± 5 Ma. The new SHRIMP age of 1619 ± 5 Ma is marginally older than, but indistinguishable from, the previously determined 1613 ± 5 Ma age (Page & Sweet 1998). The new age is interpreted as the time of zircon crystallisation and the tuff's depositional age in the lower Doomadgee Formation.

Isa Superbasin phase, southern Lawn Hill Platform

The McNamara Group, subdivided into two parts on the basis of lithostratigraphy (Hutton *et al.* 1981), dominates outcrop throughout most of the Lawn Hill Platform. The lower part comprises up to 4000 m of shallow marine to terrestrial, carbonate-dominated facies of the Torpedo Creek Quartzite, Gunpowder Creek Formation, Paradise Creek Formation, Esperanza Formation and Lady Loretta Formation. The upper part comprises up to 7500 m of deeper marine, paralic and terrestrial siliciclastic-dominated facies of the Shady Bore Quartzite, Riversleigh Siltstone, Termite Range Formation and Lawn Hill Formation. Strata thin to the north and west (where they are concealed by younger units) and are not preserved to the south and east.

Initial sequence stratigraphic studies in the NABRE project concentrated on units in the McNamara Group along the Mt Gordon Fault zone (Figure 1). This fault zone is approximately coincident with a palaeogeographic hinge zone called the Mt Gordon Arch (Derrick 1982) that defines the southeastern margin of the Lawn Hill Platform and had a long-lived influence on basin development. Several SHRIMP ages obtained from felsic tuffs in this area have a major impact on the understanding of this part of the basin's development.

GUNPOWDER CREEK FORMATION

The Gunpowder Creek Formation is a succession of predominantly thin-bedded to laminated siltstone and shale, with minor sandstone, dolostone and chert up to about 500 m thick (Southgate *et al.* 1999). Previous estimates from Hutton and Wilson (1984, 1985) of over 800 m cannot be confirmed. It conformably overlies the Torpedo Creek Quartzite, the basal unit of the McNamara Group, and includes several green to pink, K-rich beds which produce marked gamma-ray spikes. Two samples were collected from a measured section at Esperanza Waterhole (5 km west of Gunpowder township), and two others were taken from outcrops near the abandoned Mt Oxide mine (25 km along strike to the north).

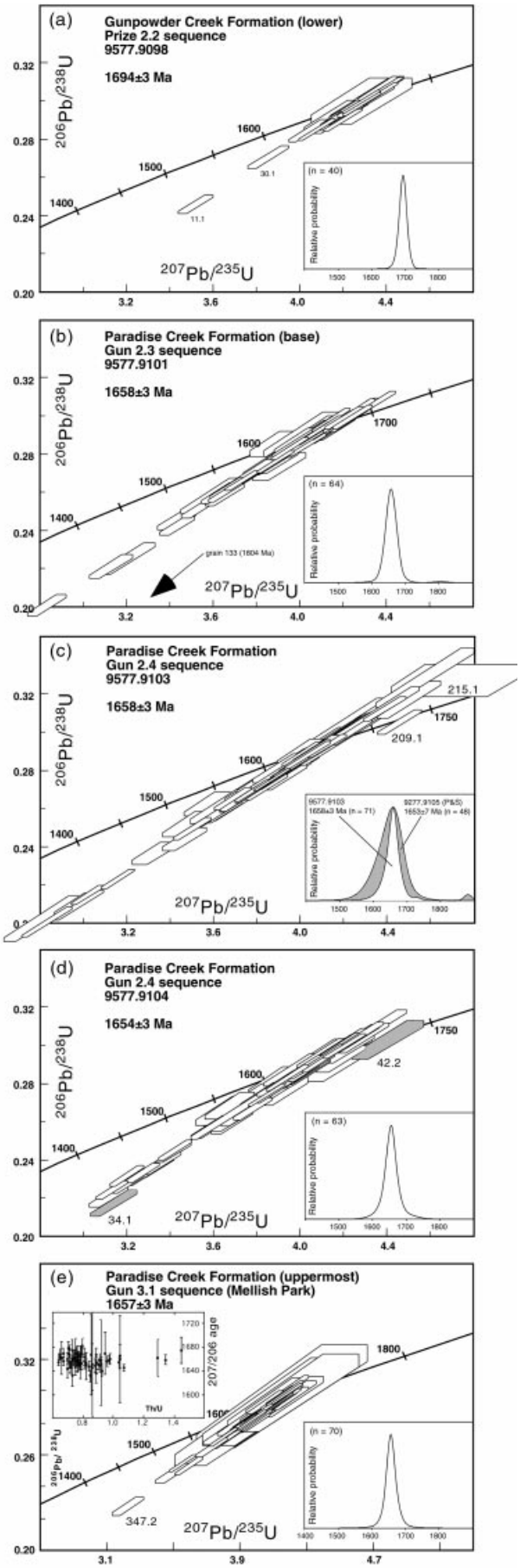
Three of the four presumed tuff layers contained either no zircon or just a few grains. The fourth (9577.9098), a 50 cm-thick pink tuff layer, has relict quartz and feldspar phenocrysts, a fine, possibly reworked groundmass, and contains abundant squat, zoned euhedral zircon. Delicately preserved pyramidal and prismatic faces on the zircon grains strongly suggest a first-cycle igneous derivation. This tuff bed occurs within a succession of interbedded intraclastic sandy dolarenite and dolomitic siltstone at the top of a shallowing-up section. It is from the lowermost part of the Gunpowder Creek Formation in the Esperanza Waterhole section, approximately 120 m above the base of the formation. The U–Pb SHRIMP results (Table 29*; Figure 11a) define a coherent group of concordant data for most of the 40 analyses from 37 zircon grains. Rejection of the two discordant analyses (11.1, 30.1) that exhibit some recent Pb loss does not alter the pooled zircon crystallisation age of 1694 ± 3 Ma (χ^2 0.79). The pristine homogeneous morphology, uniform Th/U, and unimodal age pattern render this a useful stratigraphic age for the lower Gunpowder Creek Formation.

These results have two important consequences. First, the age of 1678 ± 2 Ma from the stratigraphically concordant and underlying Carters Bore Rhyolite (Page & Sweet 1998) should now be interpreted as an intrusive age. Second, a significant time break is implied within the Gunpowder Creek Formation. When the section was first measured in 1995, sequence-stacking patterns in the upper Gunpowder Creek Formation seemed to be significantly different to those in the lower part of the formation. These changes took place across a regionally correlatable interval that contained coarser grained ferruginous clastics and carbonates, suggesting a time break of uncertain duration within the Gunpowder Creek Formation. The pre-existing SHRIMP age of 1653 ± 7 Ma (Page & Sweet 1998) from the immediately overlying Paradise Creek Formation (from the same section) was consistent with missing rock record.

PARADISE CREEK AND ESPERANZA FORMATIONS

The Paradise Creek and Esperanza Formations, a mixed succession of fine-grained clastics and carbonates with well-developed stromatolitic facies at several levels, approach a cumulative thickness of about 900 m. Sweet and Hutton (1982) and Hutton and Wilson (1984, 1985) interpreted these units as a conformable succession deposited largely in shallow-water shelf environments. Sami *et al.* (2000) have refined the setting to a regionally extensive southeast-facing mixed carbonate/siliciclastic ramp with complex interfingering facies ranging from shoreface and barrier island clastics through subtidal tempestites to deep-marine mudstones.

Several pink beds from within grey-green fine-grained dolomitic siltstone in both the Paradise Creek and Esperanza Formations were sampled in the Esperanza Waterhole section (Figure 1, locality 29). Each of these high-K beds, 5–20 cm thick, corresponded to a radiometric 'high' in outcrop. One tuffaceous layer (9577.9100) contained only rounded detrital zircons that have not been analysed. Sample 9577.9101, from 10 m above the base of the outcropping Paradise Creek Formation on the south bank of the Esperanza Waterhole section, has a dominant tuffaceous texture and a relatively homogeneous unabraded zircon suite. This is of igneous aspect and is reflected in the SHRIMP isotopic data (Table 30*; Figure 11b). Rejecting the one very discordant 1804 Ma xenocryst (grain 133) the remaining 63 concordant and discordant data points fit a single $^{207}Pb/^{206}Pb$ age distribution (χ^2 0.98) at 1658 ± 3 Ma.



The age can be interpreted as the tuff's stratigraphic age and is unchanged whether or not the four most discordant (>20% Pb loss) or 18 most discordant (>10% Pb loss) analyses are discarded.

The pre-NABRE project SHRIMP zircon age of 1653 ± 7 Ma referred to above (see Gunpowder Creek Formation) is from a tuff bed 97 m above sample 9577.9101. New SHRIMP U–Pb zircon results from an additional sample along strike in the same pinkish grey bed (9577.9103) give a concordant and discordant data array similar to that reported by Page and Sweet (1998). Replicate SHRIMP analyses (totalling 71: Table 31*; Figure 11c) were made on several of the few dozen zircon grains which were available. Grain 209.1, an obvious xenocryst (1731 Ma) can be discarded, as can 16 data points more than 10% discordant. The remaining 54 analyses have the same $^{207}Pb/^{206}Pb$ within experimental uncertainty (χ^2 0.84) resulting in a pooled age of 1658 ± 3 Ma.

A third felsic tuff bed (9577.9104) in the Paradise Creek Formation, 57 m stratigraphically higher than 9577.9103, contains a similar euhedral zircon suite. Pooling of 61 concordant and discordant data points (Table 32*; Figure 11d) gives a weighted mean $^{207}Pb/^{206}Pb$ age of 1654 ± 3 Ma (χ^2 0.85), with grains 34.1 and 42.2 having older ages of 1711 Ma and 1724 Ma, respectively. The 47 analyses that are concordant or less than 10% discordant have the same $^{207}Pb/^{206}Pb$ age of 1654 ± 3 Ma (χ^2 0.87).

In summary, the three new tuff depositional ages (1658 ± 3 Ma, 1658 ± 3 Ma, 1654 ± 3 Ma) in the Paradise Creek Formation substantiate the earlier determined, somewhat less precise age of 1653 ± 7 Ma. The internal consistency of these results provides further verification of the stratigraphic significance of the tuff ages.

Additional geochronological control on the upper part of the Paradise Creek Formation is available from a measured section at Mellish Park, 58 km north of Esperanza Waterhole. Here, a 15 cm-thick hematised pink

Figure 11 (a) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data (40 analyses) from cherty pink tuffites in the Gunpowder Creek Formation at Esperanza waterhole (9577.9098). Two numbered discordant analyses are referred to in the text. (b) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from the stratigraphically lowest pink tuffite (9577.9101) recognised in the Paradise Creek Formation at Esperanza Waterhole. The dataset of 64 analyses is unimodal in $^{207}Pb/^{206}Pb$ except for one discordant grain, 1804 Ma. (c) U–Pb concordia plot for 71 SHRIMP analyses from a cherty pink tuffite in the Paradise Creek Formation (9577.9103) at Esperanza Waterhole. Grain 209.1 (1731 Ma) and 16 analyses >10% discordant are discarded from the pooled zircon crystallisation age. The probability distribution plot (shaded portion) from Page and Sweet's (1998) zircon data in the same tuff bed (9277.9105) is overlaid on the 9577.9103 probability diagram. (d) U–Pb concordia plot and probability distribution plot of 63 SHRIMP data points from the youngest recognised pink tuffite in the Paradise Creek Formation (9577.9104) at Esperanza Waterhole. Two shaded data boxes are older analyses (1711 Ma, 1724 Ma) left out of the pooled age determination. (e) U–Pb concordia plot and probability distribution plot for 70 SHRIMP zircon analyses from a pink tuffite in the uppermost Paradise Creek Formation at Mellish Park. All except three analyses have relatively homogeneous Th/U, consistent with the uniform age distribution at 1657 ± 3 Ma.

tuff (9577.9106) is located within a section of sandy dolarenite and dololutite, 8 m below the contact between the Paradise Creek and Esperanza Formations (Southgate *et al.* 1999). The rock has angular quartz and feldspar grains suspended in a fine groundmass; it includes a strongly euhedrally zoned igneous zircon suite. SHRIMP U–Pb data from all 70 analyses (Table 33*; Figure 11e) fit a single $^{207}Pb/^{206}Pb$ age distribution (χ^2 0.77) indicating a crystallisation age of 1657 ± 3 Ma, whether or not four discordant analyses (79–87%) are included. Deletion of the five analyses with large $^{207}Pb/^{206}Pb$ errors makes no difference to the pooled age of 1657 ± 3 Ma (χ^2 0.82). Once again, the petrography, morphology, relatively uniform zircon Th/U, and a unimodal age distribution enable this igneous crystallisation age to be interpreted as approximating tuff deposition. This tuff age is indistinguishable from the depositional ages in the older parts of the Paradise Creek Formation at Esperanza Waterhole.

The new 1694 ± 3 age from the lower Gunpowder Creek Formation and the *ca* 1655 Ma ages from the Paradise Creek Formation imply a major surface of omission (approaching 20–30 million years duration) between the base of the outcropping Paradise Creek Formation and the stratigraphic level of Gunpowder sample 9577.9098. Detailed gamma-ray logging and sequence-stratigraphic research in 1996 identified a basinward-shift in facies (from thin wavy-bedded micaceous siltstone to thick-bedded medium-grained ferruginous, sandy, intraclast grainstone) approximately 50 m above the level at which the Gunpowder Creek Formation was sampled. This was initially identified as 'Surface B' in Southgate *et al.* (1997). In further studies, Sami *et al.* (2000) and Southgate *et al.* (2000) have documented the character of this erosion surface, which is mantled with conglomerates near Crocodile Waterhole (24 km to the south) and involves extensive erosion of the underlying lower Gunpowder Creek Formation near Mellish Park; this is now referred to as the Gun Supersequence and Isa Superbasin boundary. This stratigraphic break is also marked by a major magmatic event including the intrusion of the Sybella Granite and Carters Bore Rhyolite. Part of the Sybella Granite is about 1670 Ma (Page & Bell 1986), close to the Carters Bore Rhyolite age of 1678 ± 2 Ma.

LADY LORETTA FORMATION

The Lady Loretta Formation comprises up to 2000 m of laminated, stromatolitic, and intraclastic dolostone, dolomitic siltstone, dolomitic sandstone, and a basal ferruginous breccia (Sweet & Hutton 1982; Dunster & McConachie 1998). Prior to the NABRE project, several attempts were made to date possible tuffaceous horizons in this formation in the vicinity of the Lady Loretta deposit, but none were successful. The samples were either devoid of zircon or were entirely dominated by clastic zircon components. In this study, nine samples from the Lady Loretta Formation, mostly tuffites or grey-green shales which gave high gamma-ray responses, were analysed. Eight were collected from near the Lady Loretta mine, the ninth from the Kamarga Dome, 120 km to the north-northwest. Three yield depositional ages that are stratigraphically useful.

Sample 9577.9090 from above the mineralised section at the Lady Loretta deposit yielded 45 zircon grains, half of which are abraded and clearly detrital in origin, and the other half are euhedral zircons of igneous aspect. The subrounded grains form older groups with a variety of detrital ages around 2820, 1840, 1780, 1720–1700 and 1685 Ma

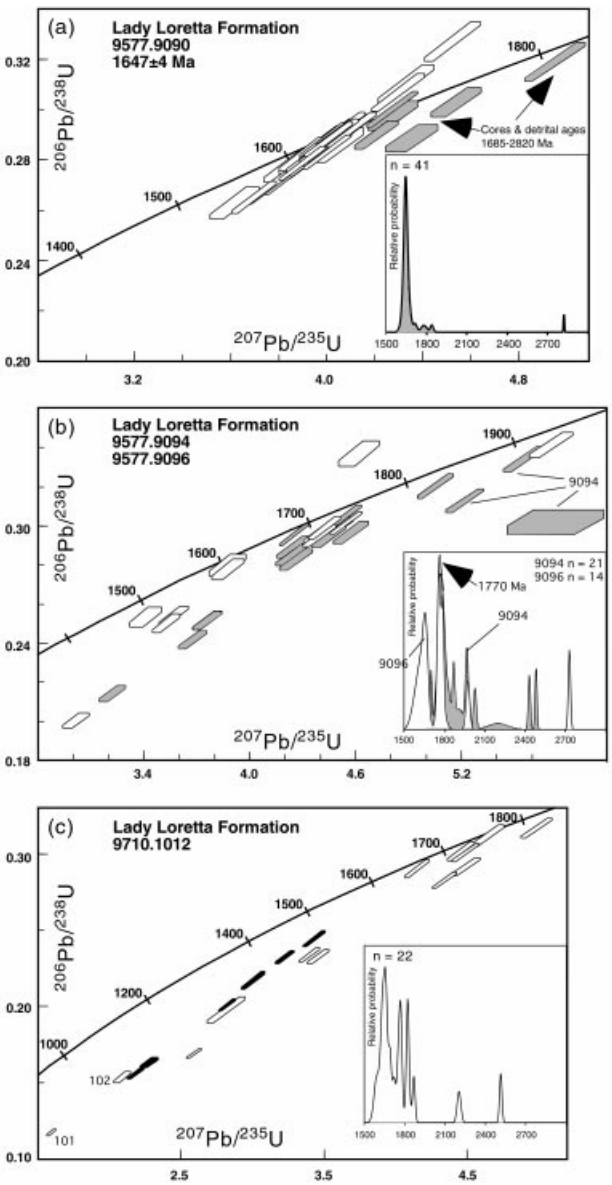


Figure 12 (a) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a tuffaceous sediment in the Lady Loretta Formation (9577.9090). Shaded error boxes are data from abraded detrital zircons of 2820–1685 Ma; the 34 open error boxes define the zircon crystallisation age of the dominant suite. (b) U–Pb concordia plot and probability distribution plots of zircon SHRIMP data from two tuffaceous sediment in the Lady Loretta Formation (9577.9094, shaded error boxes; 9577.9096, open error boxes). Provenance age patterns are similar in both sediments, especially the prominent *ca* 1770 Ma peak. (c) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a tuffaceous sediment in the Lady Loretta Formation at Kamarga Dome (9710.1012). The discordant data indicate a range of apparent $^{207}Pb/^{206}Pb$ ages, being detrital populations at 1770, 1820, 1870, 2200 and 2510 Ma, and an ill-defined discordant younger group (black error boxes) at 1648 ± 13 Ma.

(Table 34*; Figure 12a). SHRIMP data from the 34 analyses of more pristine, unabraded zircons (all with relatively uniform Th/U) form a generally concordant and experimentally coherent cluster in $^{207}Pb/^{206}Pb$ (χ^2 0.89) and define an age of 1647 ± 4 Ma. Given the reworked nature of this tuff, this is a maximum age for the Lady Loretta Formation. It is, however, stratigraphically consistent with the ages of the surrounding sequences (especially the maxima determined by the 1659–1653 Ma ages from the older Paradise Creek Formation) and therefore 1647 ± 4 Ma can be interpreted as a valid depositional age.

Samples 9577.9094 and 9577.9096 were located below the Lady Loretta ore horizon. Both contain only rounded zircon populations consisting of a dominant *ca* 1770 Ma detrital suite and a minor *ca* 1970 Ma suite, with variable amounts of other Palaeoproterozoic and late Archaean detritus. Sample 9096 also has a diffusely defined *ca* 1650 Ma zircon suite (?time of deposition). The general pattern of the provenance signatures in all three Lady Loretta rocks is similar (Table 35*; Figure 12b).

A sample from near the Kamarga Dome (9710.1012) yielded plentiful zircon. Although euhedral, most zircons are strongly zoned and altered and the data interpretation is plagued by substantial Pb loss from most of the zircon population. The clearly older detrital groups in this rock have a spread of variably discordant ages (prominent at 1870, 1820 and 1770 Ma) similar to those found in the other Lady Loretta rocks (Table 36*; Figure 12c). Two grains (101, 102) have young apparent $^{207}Pb/^{206}Pb$ ages close to 1600 Ma, but their high common Pb and discordance render them uninterpretable. Seven other zircons in this rock form an ill-defined, variably discordant population (1660–1630 Ma age range) that has a weighted mean $^{207}Pb/^{206}Pb$ age of 1648 ± 13 Ma (χ^2 0.80). Clearly, this is in good agreement with the age of the 1647 ± 4 Ma tuffaceous sediment near Lady Loretta, but this Kamarga result is not an independently acceptable age as the U-Pb data are few and affected by severe recent Pb loss.

RIVERSLEIGH SILTSTONE

The Riversleigh Siltstone is exposed in a series of domes and basins over much of the central Lawn Hill Platform (Sweet & Hutton 1982). The dominant facies are quartz siltstone, dolomitic siltstone and carbonaceous siltstone, with a maximum thickness of around 3000 m. The formation forms the bulk of the River Supersequence that occurs as a north-thinning wedge on the central Lawn Hill Platform (Krassay *et al.* 2000a). The base of the formation is a sharp contact with underlying sandstone-dominated facies of the Shady Bore Quartzite. The upper contact with the Termite Range Formation is also sharp and appears to be erosional. On regional cross-sections the Termite Range Formation abruptly overlies different Riversleigh Siltstone facies in different areas and seismic sections to the north show regional truncation below this surface (Bradshaw & Scott 1999).

Sweet and Hutton (1982) subdivided the Riversleigh Siltstone into a siltstone and shale member (Pmr), intervening sandstone bodies ($Pmrs_1$, $Pmrs_2$, and $Pmrs_3$) and an upper carbonaceous siltstone member (Pmr_w). Andrews (1998) further subdivided the existing members on the basis of detailed mapping surrounding the Century mine area. The three mapped sandstone members are important for correlating cycles within the Riversleigh Siltstone across the Lawn Hill Platform, but Krassay *et al.* (2000a) showed that there are, in fact, many more sand bodies in the Riversleigh Siltstone, many of which lack regional significance. A wide range of palaeoenvironments is interpreted for facies of the Riversleigh Siltstone. Sweet and Hutton (1982) suggested a tidal-flat origin for $Pmrs_1$ sandstone, a probable deeper water setting for $Pmrs_2$ sandstone, and either a shallow or deep-water channel setting for $Pmrs_3$ sandstone. Siltstone in the lower part of the formation was most probably deposited in subtidal marine settings in the range of storm and fairweather current activity. Upper, more carbonaceous and finer grained parts of the Riversleigh Siltstone suggest deeper water, euxinic environments (Sweet & Hutton 1982).

Numerous Riversleigh Siltstone samples were collected for dating, but only a few yielded reasonable zircon populations. Compared to correlative River Supersequence strata in the McArthur Basin, the Riversleigh Siltstone has a paucity of felsic tuffs, perhaps indicating a much higher siliciclastic sedimentation rate south of the Murphy Inlier. Three of these samples provide provenance information only, and two provide stratigraphically useful depositional ages.

Three SHRIMP samples collected from measured sections on the southwestern side of the Termite Range Fault (Figure 1) are important because they provide some constraints on the age of stratigraphic growth within the main depocentres during early River Supersequence time. They occur in fine-grained parts of the lower Riversleigh Siltstone, which was deposited in local half-graben depocentres created by growth faulting (Krassay *et al.* 2000a). Sample (9677.9127), from the Grevillea South measured section (Krassay *et al.* 1999), is 22 m above the upper contact of the Shady Bore Quartzite. This sample yielded only two detrital zircon grains, and these have apparent $^{207}Pb/^{206}Pb$ ages of 1623 ± 13 and 1755 ± 12 Ma (Table 37*). These would fit into any of the other detrital zircon age suites, so alone are not interpretable.

Sample 9677.9128, collected from the lower Riversleigh Siltstone at Grevillea South 90 m above the Shady Bore Quartzite, is a pale-green tuffaceous siltstone. It yielded a small number of homogeneous euhedrally zoned zircons of probable igneous origin, and a few abraded detrital-looking grains (grains 8 and 2) with ages of 1769 Ma and 2037 Ma, clearly older than the main suite. Inclusion-free, low common Pb domains are uncommon and most of the data falls off-concordia in a modern Pb loss pattern (Table 37*; Figure 13a). However, the 11 SHRIMP analyses give a single age cluster indicating a weighted mean $^{207}Pb/^{206}Pb$ age of 1647 ± 8 Ma (χ^2 0.62). Although the age is relatively imprecise it is consistent with geological expectations and is interpreted as a reasonable age for deposition in this part of the Riversleigh Siltstone.

Approximately 5.5 km south of Grevillea South, another lower Riversleigh Siltstone sample was collected from about 25 m above the $Pmrs_1$ sand body in a separate fault-bounded depocentre at Twin Hills (Krassay *et al.* 1999). Here, a pink, fine-grained sample (9577.9089) contains a

heterogeneous zircon population of rounded, subrounded, and euhedral grains. Although two grains of the subconcordant SHRIMP dataset (Table 38*; Figure 13b) have relatively young ages (1688 ± 10 Ma, 1671 ± 23 Ma) the major detrital suite reflects a provenance terrain about 1770 Ma, with minor provenance input from Barramundi-age (1855 Ma) and late Archaean terrains (2475 Ma). The *ca* 1770 Ma provenance signature was also evident in both of the previous Riversleigh Siltstone samples.

From the Police Creek 1 section, 16 km east of Twin Hills and not connected to small half-graben depocentres (Krassay *et al.* 1999) a laminated pinkish tuffaceous siltstone sample (9677.9133) probably represents the same stratigraphic interval as sample 9577.9089. Sample 9677.9133 contains suspended angular crystals of quartz and feldspar in a finely layered 'groundmass', and a euhedral unabraded zircon population. Most of the zircons are strongly zoned and selectively altered and this gives rise to high common Pb corrections which propagate larger than usual age errors in the U–Pb SHRIMP data (Table 39*; Figure 13c). Nevertheless, the data are mostly concordant, and all 27 analyses (except grain 4 which is 1859 Ma) define a $^{207}Pb/^{206}Pb$ age of 1663 ± 9 Ma (χ^2 0.84). When the five more discordant grains are discarded (including two of the oldest, 211.1, 3.1) the preferred age given by the remaining 21 analyses (>95% concordant) is effectively unchanged at 1662 ± 9 Ma. Like any U–Pb zircon age from tuffaceous sediment, this is strictly a maximum age for the stratigraphy. It represents a problem because it is some 15 million years older than sample 9128 from the lower Riversleigh (1647 ± 8 Ma) and older than the age of the underlying Lady Loretta Formation. The large errors, however, mask any firm conclusions and it is possible that the 1662 ± 9 Ma age population is composed of older and younger age groups which cannot be discerned due to the large individual age errors, but we cannot substantiate this explanation statistically or with any zircon morphological correlation. There is no concern about $^{207}Pb/^{206}Pb$ instrumental accuracy, as this sample was analysed concurrently with 9610.0501 (which gives a predictable Riversleigh Siltstone age, see below) and 19 analyses of the joint QGNG monitor provided a predictable $^{207}Pb/^{206}Pb$ age of 1848 ± 4 Ma. The Police Creek 1 Riversleigh Siltstone may be a stratigraphic unit older than hitherto thought, but this is considered highly unlikely (see correlations in Krassay *et al.* 2000a). It appears likely therefore that the entire 1662 ± 9 Ma zircon population in 9677.9133 is derived from a proximal igneous source that was uplifted and shedding detritus at Riversleigh Siltstone time.

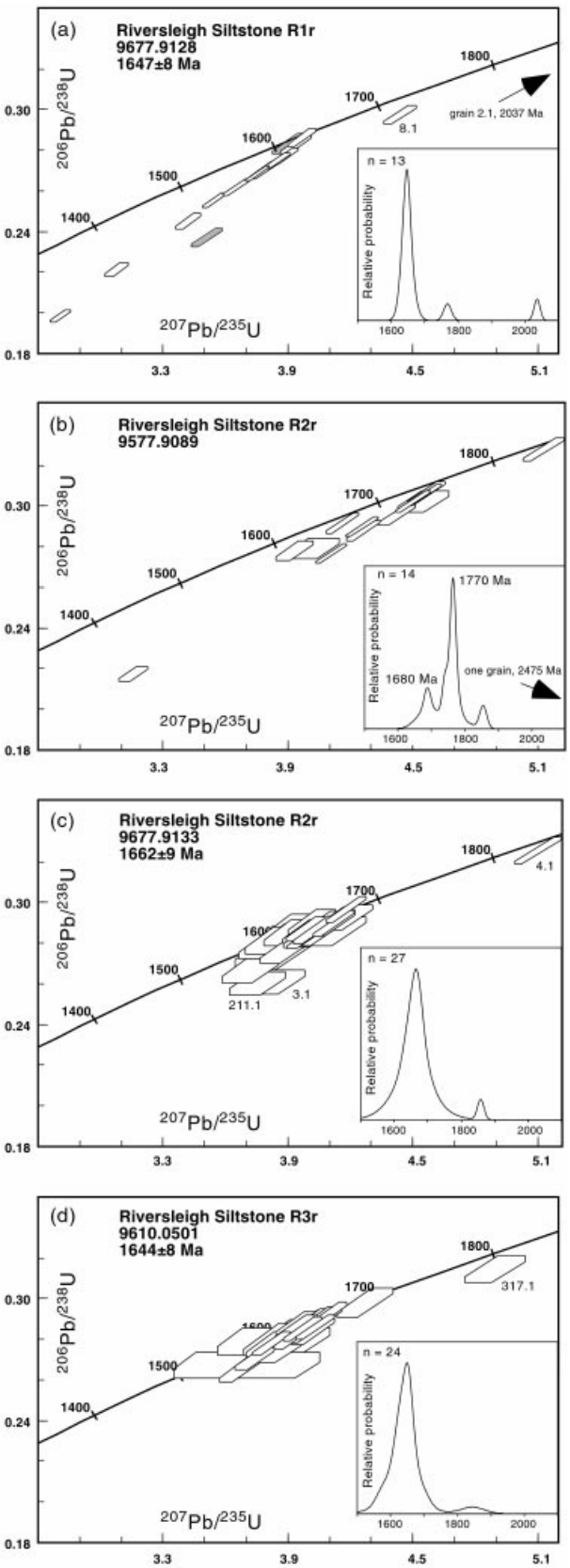


Figure 13 (a) U–Pb concordia plot of zircon SHRIMP data from tuffaceous sediments in the Riversleigh Siltstone unit R1r (9677.9128, open error boxes; 9677.9127, two shaded error boxes) and probability age distribution plot for sample 9677.9128. From the latter sample, 11 of the 13 variably discordant data points define a single $^{207}Pb/^{206}Pb$ age of 1647 ± 8 Ma. (b) U–Pb concordia plot and probability age distribution plot of zircon SHRIMP data from a sediment in the Riversleigh Siltstone unit R2r (9577.9089). The dominant detrital zircon suite is *ca* 1770 Ma; analysis 7.1 (2475 Ma) not shown. (c) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a tuffaceous sediment in the Riversleigh Siltstone unit R2r near Police Creek (9677.9133). The main euhedral zircon suite defines a unimodal age of 1662 ± 9 Ma. (d) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a sediment in the middle to upper Riversleigh Siltstone unit R3r (9610.0501). The main concordant dataset provides a zircon crystallisation age of 1644 ± 8 Ma.

The final Riversleigh Siltstone sample (9610.0501) was obtained in core from drillhole DD95FT029 located approximately 17 km southeast of the Century deposit, to the northeast of the main Riversleigh Siltstone depocentres. Here, on the northeastern side of the Termite Range Fault a much thinner, dominantly fine-grained upper Riversleigh Siltstone sedimentary succession is preserved. Detailed sequence-stratigraphic correlations place the sample in a high gamma-ray, maximum flooding zone stratigraphically below distal facies of member $Pmrs_3$ (Krassay *et al.* 2000a). The sample is a tuffaceous siltstone containing about 25 clear euhedral, weakly zoned zircons. The SHRIMP data (Table 40*; Figure 13d) reveal a zircon suite with uniform Th/U that is generally concordant and has a weighted mean $^{207}Pb/^{206}Pb$ age of 1644 ± 8 Ma (χ^2 1.05). However, grain 317 is interpreted as an 1846 Ma detrital grain and a vestige of Barramundi provenance seen in other Riversleigh Siltstone samples. As found with other Riversleigh Siltstone samples, there are moderately high common Pb corrections in some analyses. Despite the large resultant individual and pooled age errors, the 1644 ± 8 Ma age can be unambiguously interpreted as the tuff's zircon crystallisation age, and hence is a good stratigraphic age for the middle to upper Riversleigh Siltstone. The age is in good agreement with that obtained in the lower Riversleigh Siltstone at 1647 ± 8 Ma, and the two results determine reasonably firm age constraints for this part of the River Supersequence.

TERMITE RANGE FORMATION

The Termite Range Formation comprises up to 950 m of thickly bedded, coarse- to fine-grained lithic sandstone, quartzwacke and interbedded siltstone and greywacke. Sweet and Hutton (1982) divided the formation into three mappable members (Pmt_1–Pmt_3) and interpreted both the upper and lower boundaries of the formation as conformable. Krassay *et al.* (2000b) present evidence that the lower contact is a major unconformity, and the upper contact is a significant transgressive surface marking a regional change in depositional systems. Andrews (1998) divided the formation into two broadly fining-upward cycles, both characterised by thick basal sandstone and upper shale. Sweet and Hutton (1982) favoured a turbidite origin for the Termite Range Formation, an interpretation confirmed by Andrews (1998), who described the formation as sand-rich, high-concentration turbidites deposited in deep subaqueous environments.

A coarse clastic feldspathic rock from unit Pmt_3 in the upper Termite Range Formation (9577.9075) was found to have a morphologically heterogeneous zircon population. The U–Pb SHRIMP data for 37 grains (Table 41*; Figure 14) reflect this complex provenance age pattern. Approximately one quarter of the analysed grains define age peaks around 2900 Ma and 2800 Ma. There are a number of ages in the range 2300 Ma, 2200 Ma and Barramundi-ages around 1860–1800 Ma. A significant number of ages are in the 1750–1740 Ma range. The youngest 'group' appears to be about 1650 Ma, but may be composed of two age groups at 1670 Ma and 1630 Ma. The data are too few to establish firmer provenance domain ages, but a maximum age of about 1630 Ma is consistent with the previously reported U–Pb SHRIMP ages for tuffaceous siltstone from the lower and upper Termite Range Formation of 1636 ± 10 Ma and 1630 ± 5 Ma (Page & Sweet 1998). These two ages provide an approximate chronology for the lower, sand-rich part of the Term Supersequence.

LAWN HILL FORMATION

The Lawn Hill Formation is intersected in numerous drillholes near the Century deposit and crops out as small ridges on black soil plains in the central Lawn Hill

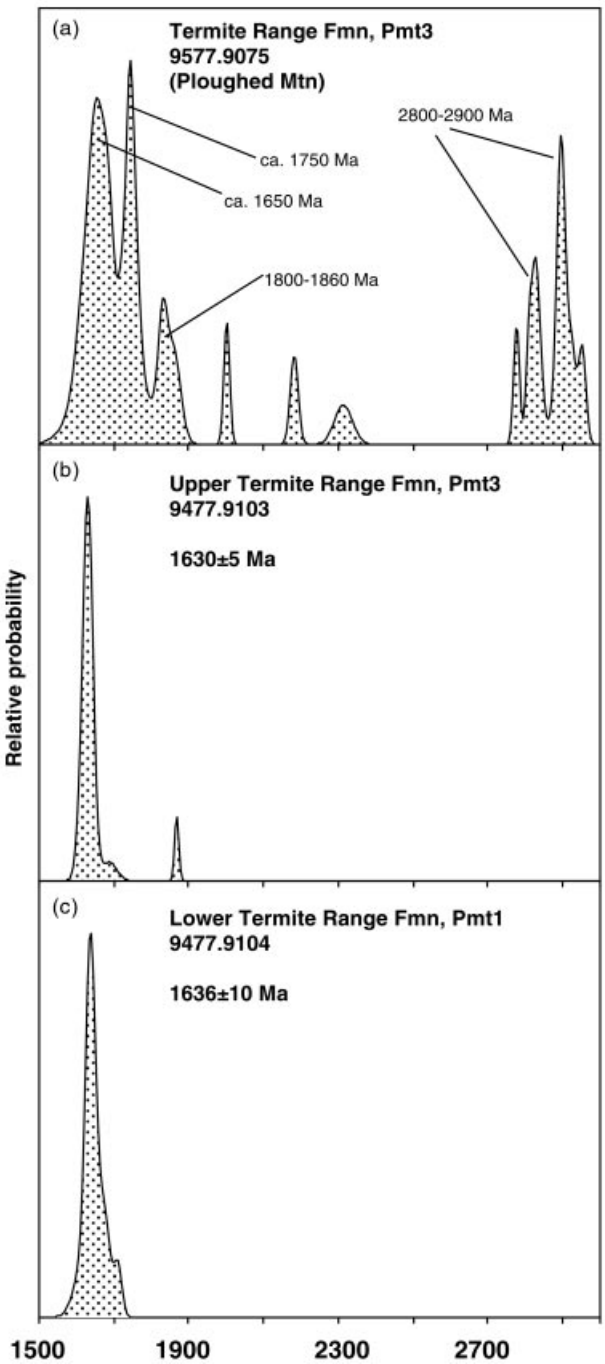


Figure 14 Probability distribution plots of zircon $^{207}Pb/^{206}Pb$ ages in clastic sediments in the Termite Range Formation (a: sample 9577.9075; b, c: data from Page and Sweet, 1998).

Platform. The Lawn Hill Formation is up to about 2300 m thick and was subdivided by Sweet and Hutton (1982) into six mappable members, Pmh_1–Pmh_6. Andrews (1998) recognised two additional mappable members: H1s within Pmh_1 and H4s (host to Century Pb–Zn–Ag mineralisation) at the top of Pmh_4. Members Pmh_1, Pmh_2, Pmh_4 and Pmh_6 are dominantly fine grained and carbonaceous and represent quiet, deep subaqueous environments. Pmh_3 is a sandy interval and represents a shallower, higher energy marine setting. Member Pmh_5 comprises thick lithic and feldspathic sandstone interpreted either as high-energy, marine shelfal deposits (Sweet & Hutton 1982) or sandy turbidites (Andrews 1998). In sequence-stratigraphic terms the Lawn Hill Formation comprises the upper part of the Term Supersequence and the Lawn, Wide and Doom Supersequences (Krassay *et al.* 2000b).

The Lawn Hill Formation is punctuated by tuffs and reworked tuffs at a number of levels. Page and Sweet (1998) measured a tuff depositional age of 1595 ± 6 Ma in the mineralised member Pmh_4 (from Century drillcore). This tuff contained minor extraneous older zircon about 1830 Ma, and Page and Sweet (1998) found that such material dominated older tuffaceous sediments (in Pmh_2) whose depositional age could not be then determined. Four new Lawn Hill Formation samples, two from member Pmh_2, one probably from member Pmh_4 and one most likely derived from member Pmh_6, help constrain the depositional ages of the Term, Lawn, Wide and Doom Supersequences. They also provide further quantitative correlation to the Fickling Group on the northern Lawn Hill Platform and farther north into the McArthur Basin.

Two grey tuffaceous samples were collected from member Pmh_2 in a measured section approximately 13 km north of Century (Figure 1, Locality 41). These samples have remnant shard textures and remnant quartz and feldspar phenocrysts. Both contain relatively pristine, euhedral, squat to elongate zoned zircons that appear to be of igneous origin. The SHRIMP U–Pb data for 29 zircon grains in sample 9577.9068 (Table 42*; Figure 15a) include one older outlier (10.1) interpreted as a xenocryst *ca* 1850 Ma old. The main igneous age population is concordant except for grain 23.1, and the 27 data points give a well-defined age of 1616 ± 5 Ma (χ^2 1.30). Sample 9577.9108 was collected in the uppermost part of member Pmh_2; this places it at the top of the Term Supersequence (Krassay *et al.* 2000b). The SHRIMP data from this sample (Table 43*; Figure 15a) are very similar to that of 9577.9068, with four older ages (1860–1840 Ma and 1760 Ma) typical of xenocrysts or sedimentary provenance domain indicated from the other Lawn Hill Formation tuffs (compare age probability histograms: Figure 15b). The principal zircon age suite forms a generally concordant to sub-concordant array. Recent Pb loss is evident in eight analyses, and the pooled $^{207}Pb/^{206}Pb$ age for 54 data points (1611 ± 4 Ma; χ^2 0.96) is not affected by rejection of these more discordant analyses. This age is in agreement with the 1616 ± 5 Ma result from the other Pmh_2 tuff and together these ages provide a good chronological constraint for this part of the Lawn Hill Formation close to the Term/Lawn Supersequence boundary (Figure 2).

These 1616 ± 5 Ma and 1611 ± 4 Ma Pmh_2 ages are stratigraphically in accord with the age of 1595 ± 6 Ma

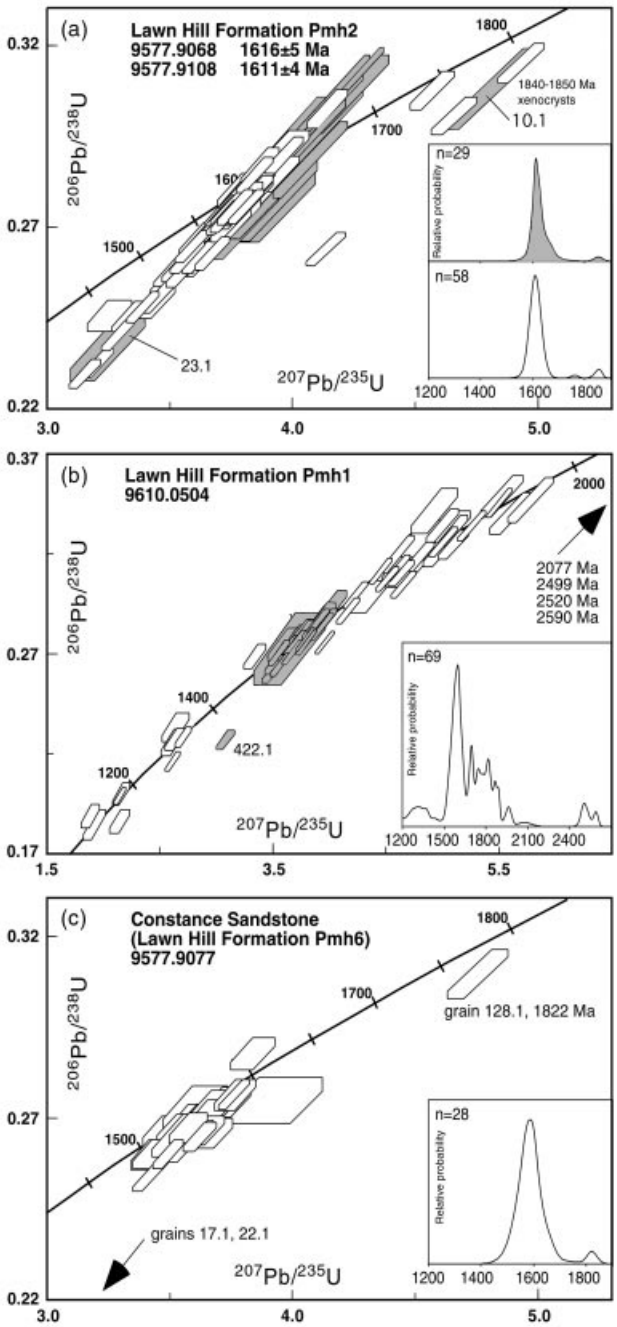


Figure 15 (a) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from tuffaceous sediments in the Lawn Hill Formation unit Pmh_2 (9577.9068, shaded error box; 9577.9108, open error box). The two zircon crystallisation ages are in accord and are considered as good stratigraphic ages for unit Pmh_2. (b) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a tuffaceous sediment in the Lawn Hill Formation unit Pmh_1 (9610.0504). The subrounded zircons have a complex pattern of provenance ages. However, neither these old ages nor the main 1593 ± 8 Ma cluster (shaded error boxes) can be unambiguously interpreted because of unequal non-zero age Pb loss, resulting in some 'ages' as young as 1000–1400 Ma. (c) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a tuff-derived sediment in the Constance Range Sandstone (9577.9077), immediately above Lawn Hill Formation unit Pmh_6. The *ca* 1600 Ma zircon suite and *ca* 1820 Ma inheritance is typical of other dated tuffs in the Lawn Hill Formation.

for the Pmh_4 tuff higher in the Lawn Hill Formation (Page & Sweet 1998). The real difference in these ages suggests that the sequence Pmh_2 to Pmh_4 (approximately the interval of the Lawn Supersequence) represents an interval of 15–20 million years. The Pmh_4 1595 ± 6 Ma tuff also includes *ca* 1830 Ma extraneous zircons, now identified in all of the dated Lawn Hill Formation rocks (Figure 15). The relevance of this provenance is amplified by the second (lowermost Pmh_4) tuff dated by Page and Sweet (1998): it contained no *ca* 1600 Ma zircons contemporary with deposition, but only 1830 Ma zircons derived from older basement (such as the Yeldham Granite). Presumably, the 1830 Ma detritus is sourced from uplifted Barramundi basement at the time of *ca* 1600 Ma Isan Orogeny.

A tuff sample (9610.0504) from drillhole DD94LH539 (5 km east of Century) occurs in black shale that is disrupted by faults and contains prominent fault breccias. The sample consists of a fine groundmass that supports isolated quartz and feldspar crystals. The majority of the zircons are abrasion-pitted and subrounded. The U–Pb SHRIMP data (Table 44*; Figure 15b) are generally concordant but are complex, indicating probable mixed terrain provenance input with ages of 2590–2500, 2080, 1960, 1890–1860, *ca* 1820, *ca* 1780, *ca* 1740, and *ca* 1680 Ma. At the other end of the analytical age spectrum there are 11 anomalously young data points from the same sample that straddle concordia between 1000 and 1400 Ma, including one duplicate analysis (411.1, 411.2). These aberrantly young analyses are difficult to interpret as they are scattered throughout the analytical session in which the standard QGNG $^{207}Pb/^{206}Pb$ age monitor was otherwise constant. As the 'young' grains are abrasion-pitted and subrounded and have no obvious morphological or geochemical differences to the rest of the population, the data can only be interpreted as having undergone significant non-zero age Pb loss. The young analyses do not represent new zircon growth as the rock is sub-greenschist facies grade and most of the grains are abraded and pitted. As this gross Pb loss effect evident in these analyses appears to be random and morphologically unpredictable, the remainder of the data in this sample cannot be confidently accepted as providing any geologically realistic ages.

The amount of subtle Pb loss that may have been experienced by the other zircon populations is unknown. For example, the most prominent age cluster (shaded in Figure 15b) is formed by 27 concordant analyses, 24 of which define a statistically coherent group (χ^2 1.24) with a pooled age of 1593 ± 8 Ma. The black shale section intersected by drillhole DD94LH539 containing the sample has been correlated to two different stratigraphic levels. Andrews (1998) correlated the section with the lower part of member Pmh_1 (Term Supersequence) while Krassay *et al.* (2000b) correlate the section with the lower part of Pmh_4 (Lawn Supersequence) on the basis of sequence stratigraphy. If the section correlates to member Pmh_4 as the latter authors believe, then the age of 1593 ± 8 Ma agrees with the 1595 ± 6 Ma age for Pmh_4 reported by Page and Sweet (1998). However, if the sample is from member Pmh_1 as proposed by Andrews (1998), then the age is in conflict with 1611 Ma and 1615 Ma ages for stratigraphically younger Pmh_2 tuffaceous rocks (described above).

Tuffaceous intervals in members Pmh_5 and Pmh_6 of the Lawn Hill Formation have been sampled but zircon yields have been inadequate. However, a sample (9577.9077) collected from immediately above member Pmh_6, 20 km due south of Century, comprises feldspathic sandstone at the base of the Constance Sandstone of the South Nicholson Group. The zircon population in this sandstone includes a few pitted, typically detrital zircons, but the majority are clear euhedral grains that have been reworked with virtually no mechanical abrasion. All except one (grain 128, 1822 Ma) of these latter zircons are isotopically uniform and SHRIMP data from 27 grains form a single $^{207}Pb/^{206}Pb$ age cluster (χ^2 1.12) at 1591 ± 10 Ma (Table 45*; Figure 15c). Grains 17.1 and 22.1 have 26% Pb loss, but they fit within the main age population. The zircon Th/U (1.0–1.5) is typically high and comparable to that of the other Lawn Hill Formation tuff zircons. The 1591 ± 10 Ma result is a maximum age for the Constance Sandstone. This is consistent with the *ca* 1500 Ma depositional age from near the base of the correlative Roper Group in the McArthur Basin (Bradshaw & Scott 1999). The 1591 ± 10 Ma sample occurs within a conglomeratic facies that contains numerous clasts of green tuffaceous siltstone. The clast type at the sample site appears similar to underlying fine-grained facies of the upper Lawn Hill Formation (Krassay *et al.* 2000b), leading us to interpret the 1591 ± 10 Ma sample as being derived from erosion of the upper Lawn Hill Formation. Thus, the age of 1591 ± 10 Ma gives a general guide to the age of this uppermost part of the Lawn Hill Formation (Figure 2).

Isa Superbasin phase, Mt Isa area

The Mt Isa Group comprises up to 2000 m of sandstone and dolomitic siltstone (Warrina Park Quartzite, Moondarra Siltstone, Breakaway Shale) overlain by dolomitic, carbonaceous, tuffaceous and siliceous siltstone and shale (Native Bee Siltstone, Urquhart Shale, Spear Siltstone, Kennedy Siltstone, Magazine Shale). Existing U–Pb SHRIMP zircon ages for the Mt Isa Group are from two tuffs in the Urquhart Shale at Isa (1652 ± 7 Ma) and Hilton (1654 ± 5 Ma) mines (Page and Sweet 1998). To substantiate the chronological tie with lower McNamara Group sequences the tuffs from the Urquhart Shale at Hilton have been reanalysed and new work carried out on two other tuffs lower in the Mt Isa Group (Moondarra Siltstone and Breakaway Shale).

MOONDARRA SILTSTONE

The Moondarra Siltstone is a succession of poorly outcropping, deeply weathered siltstone and shale approaching 800 m in thickness. A sample of K-rich shale suspected as being tuffaceous was collected at a distinctive gamma peak at 340 m in the NABRE Paroo Range section (Southgate *et al.* 1999). Because this sample is located only 40 m above the base of the formation it would give a useful age for the older part of the unit. This 8 cm-thick layer (9677.9124) contains a mixed zircon population that is mostly subrounded, but it includes some euhedral, prismatic grains. Predictably, the 47 SHRIMP data points reflect a variety of provenance source ages (Table 46*; Figure 16a). From the concordant to sub-concordant data, there is a late Archaean group around 2580 Ma, a minor 2130 Ma group, and groups at 1870 Ma and 1820 Ma. The majority of the zircon ages fit to

a group about 1780 Ma, and this can be ascribed to the well-known Argylla-aged provenance domain widespread in the Mt Isa Inlier. There is a distinct 100 million year gap to the next younger detrital age group. This suite is identified from only four grains but a number of replicate SHRIMP analyses confirm their importance in providing the best maximum age for the Moondarra Siltstone. U–Pb SHRIMP data from these four euhedral zoned and unzoned grains (112, 132, 135, 137) indicate a clearly younger age cluster; half of the zircons in this cluster have higher U than the other older zircons (Figure 16a inset). Two of these 12 young analyses are discordant and two others (135.2, 112.5) do not conform to the single age population at 1668 ± 8 Ma (χ^2 1.34) defined by the remaining eight concordant data points.

The coherence of the data at 1668 ± 8 Ma and the relatively pristine nature of these zircons suggest that this is a maximum depositional age for the Moondarra Siltstone, perhaps reflecting provenance from a *ca* 1670 Ma Sybella Granite terrain. A minimum age for the Moondarra Siltstone is given by the well-defined age of the overlying Breakaway Shale (1663 ± 3 Ma, see below). As a result, the actual depositional age for the Moondarra Siltstone is now closely bracketed within the interval 1668 to 1663 Ma. As there are at least one and a half 3rd-order sequences below the level of the dated Moondarra sample, the initiation of sedimentation in the Gun Superseqence at Mt Isa is taken at about 1670 Ma. Using composite gamma-ray curves, Southgate *et al.* (2000) correlate the Moondarra Siltstone with the upper part of the Gunpowder Creek Formation.

BREAKAWAY SHALE

The Breakaway Shale is a 200–300 m-thick unit of more resistant, laminated mudstone that outcrops more extensively than the underlying Moondarra and overlying Native Bee Siltstones. Banded pink-grey pyritic siltstone at 1110 m in the Paroo Range section (Figure 1, locality 44) was sampled (9677.9125) as a potential tuffaceous unit as it produced a prominent gamma-ray kick. This site is approximately 20 m above the base of the formation (Southgate *et al.* 1999). In thin-section, a texture consistent with a volcaniclastic origin is seen: fine wispy groundmass of quartz, feldspar and white mica supporting larger crystals of quartz and feldspar. The rock contains abundant euhedral zircon grains with fine igneous zonation and both needle-shaped (?apatite) inclusions and bleb-like inclusions typical of zircon textures in felsic volcanics. Two of three clearly older analyses—1870, 1790 and 1750 Ma, (254, 235, 242, respectively)—are from subrounded and/or pitted detrital morphologies (Table 47*; Figure 16b). These reflect similar provenance terrain to those seen in the underlying Moondarra Siltstone. The rest of the 85 SHRIMP analyses show a wide U dispersion (100–300 ppm), generally uniform Th/U (0.5–0.9), and well-grouped $^{207}Pb/^{206}Pb$ (χ^2 1.00) giving a pooled zircon crystallisation age of 1663 ± 3 Ma. Replicate analyses within grains 219 and 249 show good agreement. These data are concordant except for grains 214 and 269 (14–24% Pb loss); if the latter are discarded from the pooled data, it makes no difference to the final age. Given the isotopic coherence of the euhedral (magmatic) zircons, we can interpret the tuff crystallisation age at 1663 ± 3 Ma. This firm age for Breakaway Shale deposition is in

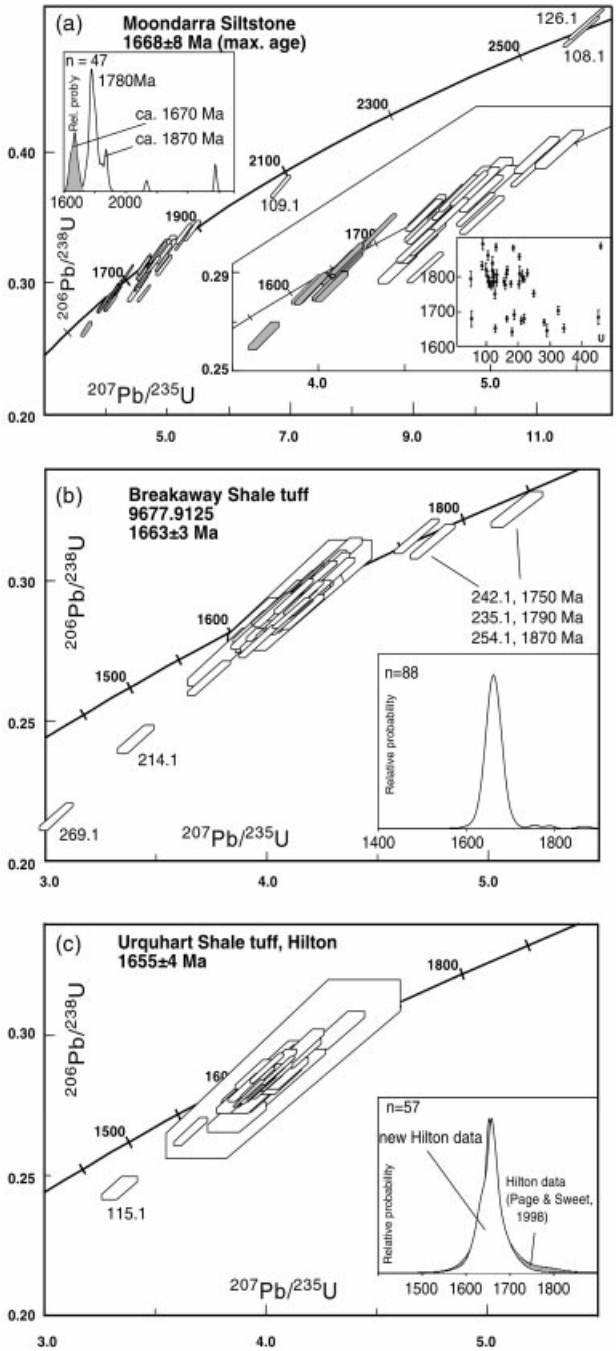


Figure 16 (a) U–Pb concordia plot and probability distribution of zircon SHRIMP data from a tuffaceous sediment in the Moondarra Siltstone, Mt Isa Group (9677.9124). Principal detrital age populations are *ca* 1870 Ma and 1780 Ma. Shaded error boxes are data that define the youngest detrital suite (1668 ± 8 Ma) and on average these zircons have somewhat higher U [inset: U (ppm) *vs* 207/206 age]. (b) U–Pb concordia plot and relative probability age distribution of zircon SHRIMP data from a tuffaceous sediment in the Breakaway Shale, Mt Isa Group (9677.9125). Apart from a few clearly older xenocrysts and discordant analyses, the coherent $^{207}Pb/^{206}Pb$ and igneous crystallisation age of 1663 ± 3 Ma indicate a geologically meaningful depositional age for the tuff. (c) U–Pb concordia plot and probability distribution plot of zircon SHRIMP data from a 'tuff marker bed' at Hilton mine in the Urquhart Shale, Mt Isa Group (9177.9005). These new data (n = 29) overlap with Page and Sweet's (1998) previous data (n = 28) to give the tuff's pooled zircon crystallisation age of 1655 ± 4 Ma.

good accord with the ages determined for underlying (Moondarra Siltstone) and overlying (Urquhart Shale) units of the Mt Isa Group. It provides the only independently determined depositional age for the base of the Gun 2.3 Sequence.

URQUHART SHALE

Previous U–Pb SHRIMP data from tuff marker beds in the Urquhart Shale at Isa mine and Hilton mine provided depositional ages of 1652 ± 7 Ma and 1654 ± 5 Ma, respectively (Page & Sweet 1998). We have reanalysed the Hilton mine tuff marker bed (9177.9005) as a check on that earlier data and to substantiate the apparent ~8 million years age difference between these Urquhart Shale ages and the new 1663 ± 3 Ma Breakaway Shale age reported above. Some of the stubby euhedral zircons in the Hilton tuff sample have high common Pb corrections and large individual age errors, but the data are generally concordant to subconcordant (Table 48*; Figure 16c). Twenty-eight of the 29 analyses (deleting grain 115, 11% discordant) fit a single $^{207}Pb/^{206}Pb$ age population of 1656 ± 4 Ma (χ^2 1.26), in excellent agreement with Page and Sweet's (1998) earlier determined 1654 ± 5 Ma result. The combined weighted mean $^{207}Pb/^{206}Pb$ age for this Hilton tuff sample (from 55 analyses) gives an age of 1655 ± 4 Ma (χ^2 0.96). Together with sequence-stratigraphic analysis (Southgate *et al.* 2000), this establishes a correlation between the base-metal-rich Native Bee – Urquhart Shale section in the Mt Isa Valley with the Paradise Creek – Esperanza Formations on the Lawn Hill Platform (Figure 2). It also confirms about a 14 million years difference between the ages of host sediments to the major stratabound Zn–Pb–Ag mineralisation at Mt Isa (Urquhart Shale) and McArthur River (Barney Creek Formation).

OVERVIEW

New U–Pb SHRIMP results (summarised in Table 1 and Figure 2) document the geochronology of basement granitic rocks and felsic tuffs in several parts of the Isa Superbasin between Mt Isa and McArthur River. The geological consistency of the results on the tuffs and sediments complements the lithostratigraphic and sequence-stratigraphic framework. In most cases, this demonstrates that the tuff zircon crystallisation ages actually approximate and date the depositional ages. On this basis, four of the eight supersequences within the Isa Superbasin, the Gun, Loretta, River and Term Supersequences, are now reasonably well dated. The others are not so well constrained and the youngest, the Doom Superseqence, is virtually unconstrained.

A major gap in the depositional rock record of ~20 million years is recognised at the base of the Gun Supersequence and this was used by Jackson *et al.* (2000a) to differentiate between the Calvert and Isa Superbasins. This break occurs about one-third of the way up the Gunpowder Creek Formation, in about the middle of the Fish River Formation, and near the top of the Masterton Sandstone (Figure 2). It indicates that although these formations have been mapped as single units, they were not deposited as continuous depositional entities. Tuffs in the lower Mt Isa Group are dated at 1668–1663 Ma (Moondarra Siltstone) and 1663 ± 3 Ma (Breakaway Shale). Depositional ages of the younger Urquhart Shale (1655 ± 4, 1652 ± 7 Ma) are comparable to or slightly younger than the ages of several pink tuff beds in the lower McNamara Group Paradise Creek Formation (1658 ± 3, 1658 ± 3, 1657 ± 3, 1654 ± 4 Ma). This suggestion of a gradual younging from the Paradise Creek Formation to the Urquhart Shale is congruent with the sequence-stratigraphic data (Southgate *et al.* 2000), but the SHRIMP age resolution is not sufficient to independently substantiate this interpretation in the Gun Supersequence.

Southgate *et al.* (2000) have used ages around 1650 Ma from widely separated units (Tatoola Sandstone 1648 ± 3 Ma, Walford Dolomite 1649 ± 7 Ma, and the Lady Loretta Formation 1647 ± 4 Ma) to define the lithologically heterogeneous Loretta Superseqence.

Likewise, the *ca* 1640 Ma River Supersequence includes a well-dated sequence in the southern McArthur Basin (Teena Dolomite 1639 ± 6 Ma, Barney Creek Formation 1639 ± 3 Ma, Lynott Formation 1636 ± 4 Ma), near the Roper River (St Vidgeon Formation 1640 ± 4 Ma, Nagi Formation 1634 ± 4 Ma) and in the northern Lawn Hill Platform (Mt Les Siltstone 1640 ± 7 Ma, Riversleigh Siltstone 1647 ± 8 Ma, 1644 ± 8 Ma). This mineralised succession is not preserved at Mt Isa.

New ages from pink felsic tuff beds in the upper McArthur and Nathan Groups in the southern McArthur Basin assist in the identification and definition of the Term (1625 ± 2 Ma), Lawn (1614 ± 4, 1613 ± 4, 1609 ± 3 Ma) and Wide (1589 ± 3 Ma) Supersequences. South of the Murphy Inlier, in the northern Lawn Hill Platform, depositional ages in the lower Doomadgee Formation (1619 ± 5, 1613 ± 5 Ma), lower Lawn Hill Formation (1616 ± 5, 1611 ± 4 Ma) and upper Lawn Hill Formation (1595 ± 6 Ma) demonstrate good stratigraphic correlation between these younger supersequences of the Isa Superbasin.

Detrital zircon ages provide insights into the adjacent source regions during evolution of the superbasin phases. A number of provenance ages recur across the basin, especially *ca* 1720, 1770–1780, 1820–1830, 1850–1880 Ma, and a range of late Archaean provenance ages between 2400 and 2800 Ma. The 1720 Ma source rocks would include Calvert Superbasin sediments and bimodal volcanics that underlie parts of the Isa Superbasin. The 1770–1780 Ma source rocks are extensive in the central Mt Isa Inlier, and are represented by the Argylla Volcanics and Bottletree Formation. Almost all samples in the Lady Loretta and Riversleigh Formations received input from such 1770–1780 Ma sources, and vestiges of 1850–1880 Ma Barramundi-age source input are minor but ubiquitous in most provenance signatures throughout the sequences. By the time of Lawn Hill Formation deposition, an additional provenance age is 1820–1830 Ma, typical of igneous rock ages in the Pine Creek Inlier.

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APPENDIX 1 SHRIMP ANALYTICAL TECHNIQUES

The SHRIMP ion microprobe study was mainly carried out using SHRIMP II in the Research School of Earth Sciences, Australian National University, Canberra. The SHRIMP I instrument was used in one analytical session (to measure detrital zircon ages in the Fish River and Termite Range Formations). Compston *et al.* (1984) and Williams and Claesson (1987) gave detailed descriptions of the techniques for zircon U–Th–Pb analysis using SHRIMP.

Early in the study, zircon grains were mounted only with fragments of the SL13 zircon Pb/U calibration standard (572 Ma, $^{206}Pb/^{238}U$ of 0.0928). For most of the current work, the 1850 Ma QGNG zircon (separated from a quartz gabbronorite from the Lincoln Complex, Eyre Peninsula, South Australia: Daly *et al.* 1998) was used as the Pb/U calibration standard ($^{206}Pb/^{238}U$ of 0.3324). It has benefits as a SHRIMP Pb/U standard and monitor of $^{207}Pb/^{206}Pb$ measurement in any given analytical session. Weighted mean QGNG $^{207}Pb/^{206}Pb$ ages obtained in various analytical sessions from this study are summarised in Table 1.

The epoxy-resin sample disc was coated with a ~50 Å layer of Au. A 30 μm primary beam of singly negatively charged O_2 was used for the production of positively charged secondary ions. Operation at a mass resolution in excess of 6500 ensured that there were no significant spectral interferences from either simple or compound isotopic species. Isotopic ratios were measureddirectly using an electron multiplier in ion-counting mode, and no correction was made for the small amount of mass-dependent mass fractionation (~2.5‰ per mass unit) that may be induced by the sputtering process and ion extraction system.

Corrections were made for initial Pb in the zircons, using values from Cumming and Richards (1975) based on measured $^{204}Pb/^{206}Pb$. Data on standards and unknowns were collected in sets of seven scans through the species of interest. Five scans were used in SHRIMP analyses of some purely clastic detrital zircons. Data reduction used to calculate the isotopic ratios was based on counting statistics augmented by uncertainties from the degree of non-linearity of the ratioed pair; these were evaluated using software developed by Trevor Ireland (*in* Muir *et al.* 1996). Uncertainties in Pb/U include a component of uncertainty of ~1.5–2% (1σ) in the calibration curve for the standard.

Decay constants used are those recommended by the IUGS Subcommission on Geochronology (Steiger & Jäger 1977). Uncertainties in calculated ages are quoted as 95% confidence limits, although analytical errors for individual analyses in the tables and error boxes in the concordia plots are at the 1σ level. The detailed analytical dataset (Tables 2–48*) is available as Supplementary Papers and is in the Australian Geological Survey Organisation's geochronological database (OZCHRON).

SUPPLEMENTARY PAPERS

Table 2 Zircon U–Pb SHRIMP data for a porphyritic granodiorite in the Nicholson Granite, sample 9177.9056.
Table 3 Zircon U–Pb SHRIMP data for a massive granitoid in the Nicholson Granite, sample 9177.9057.
Table 4 Zircon U–Pb SHRIMP data for a felsic volcanic rock in the Cliffdale Volcanics, sample 9177.9059.
Table 5 Zircon U–Pb SHRIMP data for a porphyritic felsic volcanic within the Mt Reid Volcanics, sample 9177.9024.
Table 6 Zircon U–Pb SHRIMP data for a massive granitoid of the Urapunga Granite, sample 9177.9025.
Table 7 Zircon U–Pb SHRIMP data for a felsic volcanic in the Scrutton Volcanics, sample 2773.
Table 8 Zircon U–Pb SHRIMP data for a green tuffaceous bed in the Wollogorang Formation, sample 7977.9066.
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